CHARACTERISATION AND CLASSIFICATION OF MAJOR SOILS OF INSTITUTE VILLAGE LINKAGE PROGRAMME AREA, DODDABALLAPURA TALUK, BANGALORE RURAL DISTRICT

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Department of Soil Science and Agricultural Chemistry
UNIVERSITY OF AGRICULTURAL SCIENCES
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CHARACTERISATION AND CLASSIFICATION OF MAJOR SOILS OF INSTITUTE VILLAGE LINKAGE PROGRAMME AREA, DODDABALLAPURA TALUK, BANGALORE RURAL DISTRICT

L.SIDDESHA

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in

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CERTIFICATE

This is to certify that the thesis entitled "**CHARACTERISATION AND CLASSIFIGATION OF MAJOR SOILS OF INSTITUTE VILLAGE LINKAGE PROGRAMME AREA, DODDABALLAPURA TALUK, BANGALORE RURAL DISTRICT**" submitted in partial fulfilment of the requirements for the degree of **MASTER OF SCIENCE (AGRICULTURE)** in **SOIL SCIENCE AND AGRICULTURAL CHEMISTRY** to the University of Agricultural Sciences, Bangalore, is a record of bona-fide research work done by **MR.L.SIDDESWARA** during the period of his study in this University under my guidance and supervision and that no part of the thesis has been submitted for the award of any other degree, diploma, associateship, fellowship or other similar titles.

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INTRODUCTION

I. INTRODUCTION

Soil is a vital natural resource on whose proper use depends on the life supporting systems of a country and the socio-economic development of its people. Soils provide food, fodder and fuel for meeting the basic human and animal needs. With the growth in human and animal population, demand on soils for more food production is on the increase. However, the capacity of a soil to produce is limited and the limits to production are set by intrinsic characteristics, agro-ecological settings, use and management (FAO, 1993). This demands systematic appraisal of our soil resources with respect to their extent, distribution, characteristics, behaviour and use potential, which is very important for developing an effective land use system for augmenting agricultural production on a sustainable basis.

The 1980's became the "decade of awareness", as the productivity of Agricultural soils world-wide was in general on the decline. Per capita food grain production decreased significantly. This decline was attributed to periodic droughts, poor management and exploitative agriculture coupled with degradation process (FAO, 1986). In India, more than 60 per cent of the population depends on agriculture. Every inch of arable land has already been utilised to the maximum extent.

Since soils in general are degrading due to poor management and faulty land use at a rate, which is faster than their natural regeneration or resilience,

it becomes imperative to protect them from further degradation, as there is a concomitant decline in soil quality to produce healthy crops.

Alfisols, commonly referred to as the red soils, occupy more than 60 m.ha of land in India. Among different soil types, red soils occupy third position in extent after alluvial and black soils. Most of them are in the semi-arid tropics of southern India and some in the sub humid eastern Madhya Pradesh and Northeast India (Krishnappa *et al.*, 1998). In Karnataka 35 per cent of total geographical area is covered by red soils (Hegde *et al.*, 1987), bulk of which are in Southern Karnataka (6.6 m.ha) and in North Karnataka they are estimated to cover an area of 2.2 m.ha and are often associated with black soils.

Red soils are considered to be those which have hues of 5 YR in the soil series control section (25-200 cm) (Sehgal, 1998). Red soils are mostly formed on granites, gneisses and schists of the Archean period. Generally these soils are well drained with moderate permeability. They have light textured: sandy or gravelly to loam, with poor aggregating ability. The water holding capacity is low, they are acid to neutral in reaction, the clay fraction is dominated by kaolinite and hydrous oxides of iron and aluminium. CEC is low 10 to 15 c mol (+) kg^{-1}. The soils are low in organic matter, nitrogen, phosphorus and other plant nutrients, with the exception of potassium (Krishnappa *et al.*, 1998). In general, these soils are agriculturally important having good soil-air-water relationships and retain moisture under proper conservation practices. However, the shallow and gravelly soils are susceptible to erosion and have low water holding capacity.

The inherent ability of soil to supply nutrients for crop growth and the maintenance of soil physical conditions to optimise crop yield responses are most important components of soil fertility that virtually determines the productivity of agricultural system. A thorough and proper understanding of the morphology and physico-chemical characteristics of the soil will give greater insight of the dynamics of a given soil. By characterising the soil one can clearly understand and ascertain the inherent capacity of the soil for crop production as well as various other problems that are associated and often confronted during the management of a given soil.

Keeping this in view this study was undertaken to identify the major land forms and dominant soil resources occurring in Neralaghatta and Kestur villages of Doddaballapura taluk (Bangalore rural district) and to assess their potential for sustainable crop production.

The main objectives of this study are as follows:

1. Characterisation of major soils occurring in the Institute Village Linkage Programme Area, Doddaballapura Taluk, Bangalore Rural District.
2. To classify the soils occurring in the Institute·Village Linkage Programme Area as per soil taxonomy.
3. To identify the problems and potentials of the study area.

REVIEW OF LITERATURE

II. REVIEW OF LITERATURE

The rational utilization, improvement and conservation of soil resources are not possible without enlarging thorough understanding of their physico-chemical properties, geographical distribution and classification. Although the soils of much of the humid tropics are superficially similar in appearance, careful study reveals a great diversity. Studies carried out, both within and outside India, on red soils is not exhaustive and there is need for detailed investigation. Based on the available information the literature is reviewed under the following headings.

2.1 Morphological properties.

2.2 Physical properties.

2.3 Chemical properties

2.4 Classification.

2.1 Morphological characteristics of soils

Esu and Ojanuga (1985) while studying morphological, physical and chemical characteristics of Alfisols occurring in Nigeria observed that soil texture varied from sandy loam to gravely clay loam in the sub soil with high silt fractions and relatively low in clay content in the surface soil.

Arun Prasad *et al.* (1989) observed while relating profile development with the topography that soils on the hill top and mid slope were reddish brown to dark reddish brown, while the soils occurring on the toe slopes

and depression were dark brown to very dark greyish brown in colour. Texture of the soils of the slopes was gravely loam, while mid slope and toe slope soils exhibited clay loam to clay type of texture. Soils developed on hill slope exhibited granular structure due to high organic matter content while that of mid slope and toe slope soils were subangular blocky to angular blocky.

Morphological study of sub humid temperate high lands of Central Himalaya by Kaista and Gupta (1993) revealed that texture of soil remained sandy loam through out the depth and soils exhibited angular blocky to subangular blocky structure.

Reddy *et al.* (1993) studied morpholigical and physico-chemical properties of red soils (Alfisols) occurring under irrigated and unirrigated conditions and observed that texture of soils ranged from sandy loam on the surface to sandy clay loam in the sub soil. Colour of the Bt horizon in all irrigated soils was reddish brown where as in unirrigated soils; it was dark reddish brown.

A field survey of red and lateritic soils under mango (*Mangifera indica*) in Karnataka was carried out in eleven soils in which soil depth, coarse fragments, erosion and surface runoff were the major soil-related constraints (Naidu *et al.*, 1994).

Sharma *et al.* (1996) observed while characterising soils in a toposequence over basaltic terrain of southern Rajasthan that the soils at elevated topography were shallow to moderately shallow depth, clayey to loamy skeletal

texture and yellowish brown, while at lower topography soils were deep to very deep, fine to fine loamy texture and greyish colour.

Characterisation of Barak Valley soils of Assam revealed that low land soils in general had yellow hue and chromas of many horizons are two or less indicating extensive gleying. Upland soils did not have any redoximorphic features indicating free drainage and deep ground water table. These soils usually had a dark brown surface horizon and redder subsurface horizons (Sen *et al.*, 1997).

Walia and Rao (1997) studied characteristics and elemental composition of four pedons representing four landforms of Banda plain in Uttar Pradesh. They observed that soils of piedmont plain are well drained, reddish brown with clay loam texture, while soils of recent dissected flood plain are some what excessively drained, light yellowish brown sandy clay loam and highly calcareous with lime concretions. The soils of undulating floodplain were moderately well drained, dark greyish brown, clayey with deep cracks and weak slickensides, whereas the soils of level flood plains are imperfectly drained, fine grained, brown to olive brown, silty clay, showing surface cracks and redoximorphic features.

In Greece, red soils inherited their colour from their parent materials, that were transported from the originally formed residual soils on hard limestone (Yassaoglou *et al.*, 1997).

2.2 Physical characteristics of soils

2.2.1 Distribution of soil separates

Roy and Landey (1962) working with red and lateritic soils of Mond watershed area of Raigarh district observed that clay content of soil increases with depth but stabilise at second or third horizon.

Study with soil formed on granite in an area receiving annual precipitation of 3,257 mm by Eswaran and Wang Chaw Bin (1978) in Malaysia revealed that clay content of the soil remains constant throughout the solum indicating least translocation of clay in the profile.

Rajamannar and Krishnamoorthy (1978) noticed that the clay content of the forest soils of western ghats in south India decreased with the depth of the profile.

Characterising the red and laterite soils of northern Plateau Zone of Orissa, formed on highly weathered material. Sahu *et al.* (1990) observed that the content of clay increased with the depth in all the profiles with a simultaneous decrease in sand content indicating well-drained condition. Due to illuviation of clay there was evidence of argillic horizon in the soil.

While characterising the typical pedons of hill slope soils of middle Andaman island, Nirmalya Bala and Sahu (1993) observed that the clay distribution down the profile did not follow any regular pattern.

Characterisation of Gangetic plain soils of Bihar developed in subtropical and sub humid type of climatic condition with the annual rainfall of 1200 mm revealed that soils were lighter in colour, mainly influenced by the colour of the sand fraction. The soils were sandy to sandy loam, non-sticky, non plastic and generally of single grain to massive in structure with an ochric epipedon (Diwakar and Singh, 1994).

Datta and Karmakar (1995) worked on soils of instructional-cum-Research farm of Assam and observed that the soils of the area are very pale brown to yellowish brown in colour, fine textured, subangular blocky in structure, while clay exhibits some definite pattern of distribution indicating moderate development of the soils.

Walia and Rao (1996) working with red soils of Bundelkhand region of Uttar Pradesh observed that there was substantial increase in clay content in Bt horizon supporting there by evidence of illuviation of clay.

While characterising the soils of an irrigated river flood plain in the eastern coastal region, Sahu and Mishra (1997) observed that sand and silt content in all pedons decreased from the surface downwards where as clay content gradually increased down the depth indicating pedogenic soil development in the region.

Sarma *et al.* (1997) studied the physical properties of two soils (Alfisols and Inceptisols) from different command areas of Assam and observed that bulk density was higher in Alfisols than in Inceptisols and it increased with depth in both the soils. The surface layer of both soils contained higher percentage of aggregates less than 0.25 mm when compared to subsurface layers. The clay content was found to be the single most dominant factor influencing the physical attributes of soils in the area.

2.3 Chemical properties of soils

2.3.1 Soil reaction (pH)

The soils of eastern Himalayas were extremely acidic and surface horizons were having lower pH compared to subsurface horizons (Das *et al.*, 1988).

Sharma and Qaher (1989) working with forest profiles derived from Gneiss and Schist type of parent materials revealed that in most of the soil profiles pH of the soil did not show any specific trend with the depth.

While characterizing and classifying lateritic soils of North Karnataka, Narayana Rao *et al.* (1993) reported that the pH of soil measured in water was slightly less than 6.5 and increased with depth. The pH measured in KCl was slightly lower than that measured in water.

Sivasankaran *et al.* (1993) observed that the pH of red soils varied between 3.9 and 7.0 in Western ghat region of Karnataka and remarked that rainfall was the major factor contributed to the increase in acidity in these soils.

In Vijayapura and Tyamagondlu soil series of Karnataka, pH values ranged from moderately acidic to slightly acidic and increasing trend with the depth of the profile was noticed in both the soil series (Prakash *et al.*, 1993).

In a study on red soils of Bundelkhand region of Uttar Pradesh, Walia and Rao (1996) noted increasing pH with depth, indicating the leaching of bases in the soil.

Characterisation of soils of Barak valley in Assam by Sen *et al.* (1997) revealed that the low land soils are acidic to neutral and pH of the surface soils varies from 5.3 to 5.6 and show regular increase down the profile where as the upland soils are highly leached and have low pH.

Shivaprasad *et al.* (1998) while characterising the soils of Karnataka observed that soils derived from granitic-gueissic parent material were found to be slightly acidic to near neutral in soil reaction.

Soils of Bangalore district were found to be acidic to near neutral and showed decreasing trend with the depth (Reddy *et al.*, 1993).

2.3.2 Electrical conductivity (EC)

Working with red and black soils of Andhra Pradesh, Krishnamoorthy and Govinda Rajan (1977) observed that in red soils electrical conductivity values ranged from less than 0.15 to 0.25 dSm^{-1} and showed increasing trend with depth.

Electrical conductivity of northern Karnataka soils ranged from 23.1 to 13.8 dSm^{-1} and decreased with depth (Bhadrapur and Sheshgiri Rao, 1977).

Characterising the lateritic soils of Dakshina Kannada district of Karnataka, Satish (1991) reported that electrical conductivity of these soils were very low ranging from 0.08 to 0.32 dSm^{-1} and did not show any relation with the depth of the profile.

Sivasankaran *et al.* (1993) while characterising the soils of western ghats in Karnataka observed that the electrical conductivity values ranged from 0.1 to 0.4 dSm^{-1} indicating no accumulation of salts in the soil.

While characterising the soils of the Trans-Yamuna Plain, Walia and Rao (1997) reported that the electrical conductivity of these soils were very low ranging from 0.08 to 0.4 dSm^{-1} and did not show any specific relationship with the depth indicating the low amount of soluble salts in the soil.

2.3.3 Organic Carbon (OC)

Raychaudhari *et al.* (1941) reported that organic carbon content was little in black soil compared to that of associated red soil. They also reported that the C/N ratio was higher in black soils than that of associated red soils.

Desai (1942) reported that in general red soils were devoid of free $CaCO_3$ whereas associated black soils were in general rich in free $CaCO_3$. In some cases however, free $CaCO_3$ was present in small quantity in red soils due to the semi arid climatic condition.

Krishnamoorthy and Govinda Rajan (1977) working with black and red soils of Andhra Pradesh noticed higher organic matter in second horizon which later decreased in red soils while in black soils accumulation of organic matter was noticed in fourth horizon due to the downward movement of it along with clay.

Pal *et al.* (1985) characterising three soil profiles of Darjeeling forest soils noticed a sharp decrease of organic matter with depth in two profiles while in other, organic carbon was more in second layer than in first layer but later decreased.

The organic carbon content of soils under different plantation crops in Western ghats was found to vary between one and 10 per cent and remarked that, it varied with altitude, latitude and cultural practices (Sivasankaran *et al.*, 1993).

Working with paddy soils, Gupta and Tripathi (1993) reported that generally organic matter content of soils decreases with depth in most profiles but in case of submerged soils higher amount of organic matter was noticed in the sub soil immediately below the surface horizon.

Walia and Rao (1996) working with red soils of Bundelkhand region of Uttar Pradesh noticed that the organic carbon content of soils (0.5 to 1.5%) decreased with depth, mainly due to the changes in land use.

The organic carbon content of lateritic soils (Bangalore plateau) varied between low to medium and showed decreasing trend regularly down the profile (Shivaprasad *et al.*, 1998).

2.3.4 Cation exchange capacity and exchangeable cations

Decrease in cation exchange capacity, inspite of an increase in clay content with the depth in the soil profile in Himachal Pradesh was attributed to the higher content of organic matter in the upper horizon of the profile (Sharma and Rao, 1957).

Raghu Mohan (1962) studied the different soil mechanical separates for their contribution to CEC. It has been observed that in red and laterite soils even the coarser fractions like sand and secondary gravel referred to as concretions contributed towards CEC of soil in a significant way. This is mainly due to the pseudo nature of the concretions which had both sesquioxides and colloids embedded in the clay matrix.

Prasad *et al.* (1977) noted that cation exchange capacity of soil formed on basalt ranged from45.5 to 66.2 c.mol (p^+) kg^{-1}, exchangeable potassium showed decreasing trend with the depth while sodium indicated increasing trend and the values recorded for potassium and sodium in profile ranged from 0.3 to 0.5 c.mol (p^+) kg^{-1} and 2.0 to 4.0 c.mol (p^+) kg^{-1}, respectively. Base saturation percentage of the soil ranged from 89 to 95.1 per cent.

Walia and Chamuah (1988) studied the acidic soils in old flood plains of Assam and reported that cation exchange capacity values were low, which were attributed to predominance of kaolinite clay in the soil. Under heavy rainfall and intensive leaching, variation in cation exchange capacity values with depth in both the profiles was related to clay content.

Ragu Mohan (1989) in his study of soils of Goa has reported a significant contribution of CEC in the amorphous materials of ferralitic soils which helps to maintain nutrient sustenance as an alternative for low CEC values of 1 : 1 type clay present in soils.

Sahu *et al.* (1990) while studying soils of Northern plateau zone of Orissa observed that increase in cation exchange capacity with depth which was attributed to gradual increase in clay content with the depth of profile.

While characterising the various pedochemical characteristic of Mizoram soil, Singh *et al.* (1991) observed high cation exchange capacity in the surface soils layer compared to subsurface soils. Low values of cation exchange capacity in sub surface were attributed to the presence of pH dependent exchange sites occupied by either hydrogen or hydroxy aluminium ions at low pH. Exchangeable calcium, magnesium and potassium decreased with the depth at all altitudes while exchangeable sodium did not show such depth wise distribution.

Das *et al.* (1992) reported that cation exchange capacity of laterite soils of Orissa varied from 2.9 to 13.4 c.mol (p^+) kg^{-1} and increased with depth. Base saturation percentage of these soil ranged from 34.82 to 80 per cent.

Shrikanth and Bapat (1993) compared the cation exchange capacity of soils formed from different parent materials and observed that cation exchange capacity of soil formed from sand stone ranged from 24.3 to 25.8 c.mol (p^+) kg^{-1} and slightly increased with depth while that of soil formed on basalt decreased

from 53.8 c.mol (p^+) kg^{-1} in surface to 51.0 c.mol (p^+) kg^{-1} in subsurface layers. Alluvial soils recorded the cation exchange capacity of 38.1 to 56.7 c.mol (p^+) kg^{-1}. Exchangeable calcium remained constant throughout the depth with the values of 11.0 and 37.38 c.mol (p^+) kg^{-1} for soil derived from sand stone and basalt respectively where as exchangeable magnesium recorded the value of 4 c.mol (p^+) kg^{-1} (sand stone) and 10 to 12 c.mol (p^+) kg^{-1} (basalt). Exchangeable potassium and sodium ranged from 4 c.mol (p^+) kg^{-1} and 1.6 to 1.9 c.mol (p^+) kg^{-1} respectively for sand stone and 0.5 to 0.8 c.mol (p^+) kg^{-1} and 1.3 to 1.8 c.mol (p^+) kg^{-1}, respectively for basalt.

While characterising the red soils of Manipur, Sen *et al.* (1993) observed that the cation exchange capacity and base saturation values showed decreasing trends with the depth.

Cation exchange capacity values showed an increasing trend with the depth of the profile, whereas, per cent base saturation values decreased regularly down the profile in red and lateritic soils of southern India (Krishnan, 1993).

While characterising the lateritic soils of North Karnataka, Narayana Rao *et al.* (1993) observed that both cation exchange capacity and per cent base saturation values showed regular increasing trend down the profile.

Venkatesh and Satyanarayana (1994) noted that cation exchange capacity of Vertisols, of Karnataka ranged from 56.0 to 77.8 c.mol (p^+) kg^{-1} and [illegible] increased upto the last horizon.

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Four soil profiles representing two typical soil series of Madhupura clay in Bangladesh showed low base saturation where as soluble sodium and potassium were leached out of the soil and calcium and magnesium were the dominant cations in the profiles which increases with increasing depth (Elahi *et al.*, 1996).

Sen *et al.* (1997) while characterising some of the soils of Barak valley in Assam observed that cation exchange capacity of the soils was generally low and it varied from 6 to 21 c.mol (p^+) kg^{-1}. The CEC of surface soils were high due to higher organic matter content.

2.3.5 Available nutrients

2.3.5.1 Available nitrogen

The available nitrogen content of the surface soil ranged from 0.019 to 0.029, 0.020 and 0.014 to 0.016 per cent in Peelamedu, Periyanayakkanpalyam and Dasarpatty series, respectively and decreased with depth (Parasuraman and Jayaraj, 1982).

Mruthunjaya (1991) while studying some salt affected soils of Cauvery and Vanivilas Sagar command areas of Karnataka reported that the available nitrogen in the profiles ranged from 30.71 to 105.1 kg per hectare. Nitrogen content decreased with depth.

In Karnataka about 10.3 per cent of soils fall under low category , 35.8 percent under medium and 53.9 per cent under high category of available nitrogen status (Shivaprasad *et al.*, 1998).

2.3.5.2 Available phosphorus

Gangappa (1989) reported that the available phosphorus content of salt affected soils of Chitradurga district (Karnataka), varied from 2.69 to 80.64 kg per hectare. In profile samples, it ranged from 2.72 to 41.2 kg per hectare. The available phosphorus increased with depth in profile samples.

While characterising and classifying soils of an irrigated river flood plain in the eastern coastal region, Sahu and Mishra (1997) reported that available phosphorus content varied between 12.76 to 14.0 kg per ha and was in medium level.

The data on the available phosphorus status in the soils of Karnataka showed that about 83 per cent of the soils are low in phosphorus and 17 per cent area is under medium category (Shivarparasad *et al*., 1998).

2.3.5.3 Available potassium

The available potassium content of surface soil ranged from 0.082 to 0.03, 0.015 to 0.026 and 0.031 to 0.033 per cent in Peelmedu, Periyanakkan palayam and Dasarpatty series, respectively and in general its content decreased with depth (Parasuraman and Jayaraj, 1982).

While characterising and classifying soils of an irrigated river flood plain in the eastern coastal region Sahu and Mishra (1997) reported that the available potassium content varied from 171.0 to 211.66 kg per ha and was in medium level.

The available potassium is medium to high in most of the soils of the Karnataka state except in lateritic soils of coastal plain and western ghats and in shallow red and black soils where it is low (Shivaprasad *et al*., 1998).

2.3.5.4 Available micronutrients

The available zinc ranged from 0.05 to 2.10 ppm with a mean value of 0.84 ppm in Himachal Pradesh soils and it invariably decreased with depth (Kapur and Dev, 1977).

Sakal *et al*. (1986) reported that the available Zn in the surface soil samples in sub Himilayan hill and forest region of Bihar ranged from 0.15 to 7.80 ppm.

The available Zn in the surface soil samples in old alluvial soils of Bihar ranged from 0.04 to 5.80 ppm with a mean value of 0.66 ppm (Sakal *et al*., 1988).

Badrinath *et.al*. (1987) observed that available zinc varied from 0.3 to 1.08 ppm in saline soils of Ankola of coastal Karnataka.

Katyal and Sharma (1991) reported that available zinc varied from 0.12 and 2.80 mg/kg soil in Indian soils. It was nearly one hundredth of the total zinc in Indian soils.

The data on available zinc status and available iron status of soils of Karnataka state indicated that about 79 per cent of area of soils have zinc deficiency (Shivaprasad *et al*., 1998) and 34 per cent of the soils in the state are deficient in iron.

2.4 Classification of soils

Any natural object having many variants may not be easy to classify. The multitude of characteristics involved in the variant make the grouping difficult. The soil is a typical example wherein number of variants influencing its origin is large. Therefore, in order to understand differences, similarities and relationships among different members, it is necessary that these are grouped in some orderly manner.

Classification is the grouping of objects in some orderly and logical manner into compartments. It is based on the properties of objects for the purpose of studying, identifying and grouping them. The properties are selected in accordance with the purpose of classification. They are termed as differentiating characteristics and serve to differentiate one class from all others.

The first natural soil classification was that of Dockhuchaev of Russian school. According to him soils were independent, natural bodies possessing properties reflecting the effect of local and zonal soil forming factors. Joffe of USDA in 1912 classified soils into five classes on the basis of properties. Marbut, Baldwin, Kelley and Thorp, classified the soils based on the genesis or on the properties of so called virgin soils.

A new system of classification was presented by soil survey staff of USDA in the year 1975. According to this soils are classified into ten orders based on their properties. Each order is further divided into suborders, great groups, sub groups, families and series.

Classification of red soils of India was initiated by Raychaudhari (1961, 1962). Govindarajan and Datta Biswas (1968) classified red loamy soils as Paleustalfs, Rhodustalfs and Haplustalfs and red sandy soils as Rhodustalfs and Haplustalfs.

Govindarajan and Datta Biswas (1968) identified red soils of Muchkand basin in Korapat districts of Orissa as Latosols of high base status.

Digar *et al.* (1973) classified red and lateritic soils of Muchkund basin of Andhra Pradesh and Madhya Pradesh as Udic Haplustalfs, Ultic Paleustalfs and Ultic Rhodustalfs.

Murthy *et al.* (1982) classified red and lateritic soils of Karnataka, as Paleustalfs (Tyamagondlu series) Haplustalfs (Vijaypura series) and Rhodustalfs (Channasandra series).

Zende (1987) classified the soils of Nagaland in relation to physiography. The dominant soils are (i) fine loamy, Typic Hapludulfs on every steep side slopes of high hills (ii) loamy skeletal, Typic Udorthents on narrow ridges of high hills (iii) loamy skeletal, Typic Hapludulfs on broad ridges of hills and Paleudults on steep slopes of high hills and very low hills.

Eswaran (1988) defined red soils as "those soils that have a hue of 5YR or redder, and have a CEC of (IN NH_4OAc at pH 7) 24 meq. Per 100 g clay or less or an ECEC of 18 meq. Per 100 g clay or less. In addition, they have less than 10 per cent weatherable minerals in the fine sand fraction. They may have an

angillic or oxic horizon but donot have a Cambic horizon, and they have a thermic, isothermic or warmer iso temperature regime".

The red and lateritic soils of Bangalore district (Karnataka state) were classified as kandic Paleustalfs and Kandic Rhodustalfs by Reddy *et al.* (1993).

Five pairs of associated red and black soils occurring in close proximity were excavated in north Karnataka and soils were classified as Alfisols, Inceptisols and Vertisols (Rudramurthy *et al.*, 1996).

The soils of Karnataka belong to seven orders, 12 suborders, 27 great groups, 47 subgroups and 96 soil families. Of the total area of Karnataka 27 per cent is covered by Alfisols, 25 per cent by Inceptisols, 16 per cent by Entisols, 15 per cent by Vertisols, eight per cent by Ultisols, five per cent by Aridisols and one per cent by Mollisols (Shivaprasad *et al.*, 1998).

MATERIAL AND METHODS

III. MATERIALS AND METHODS

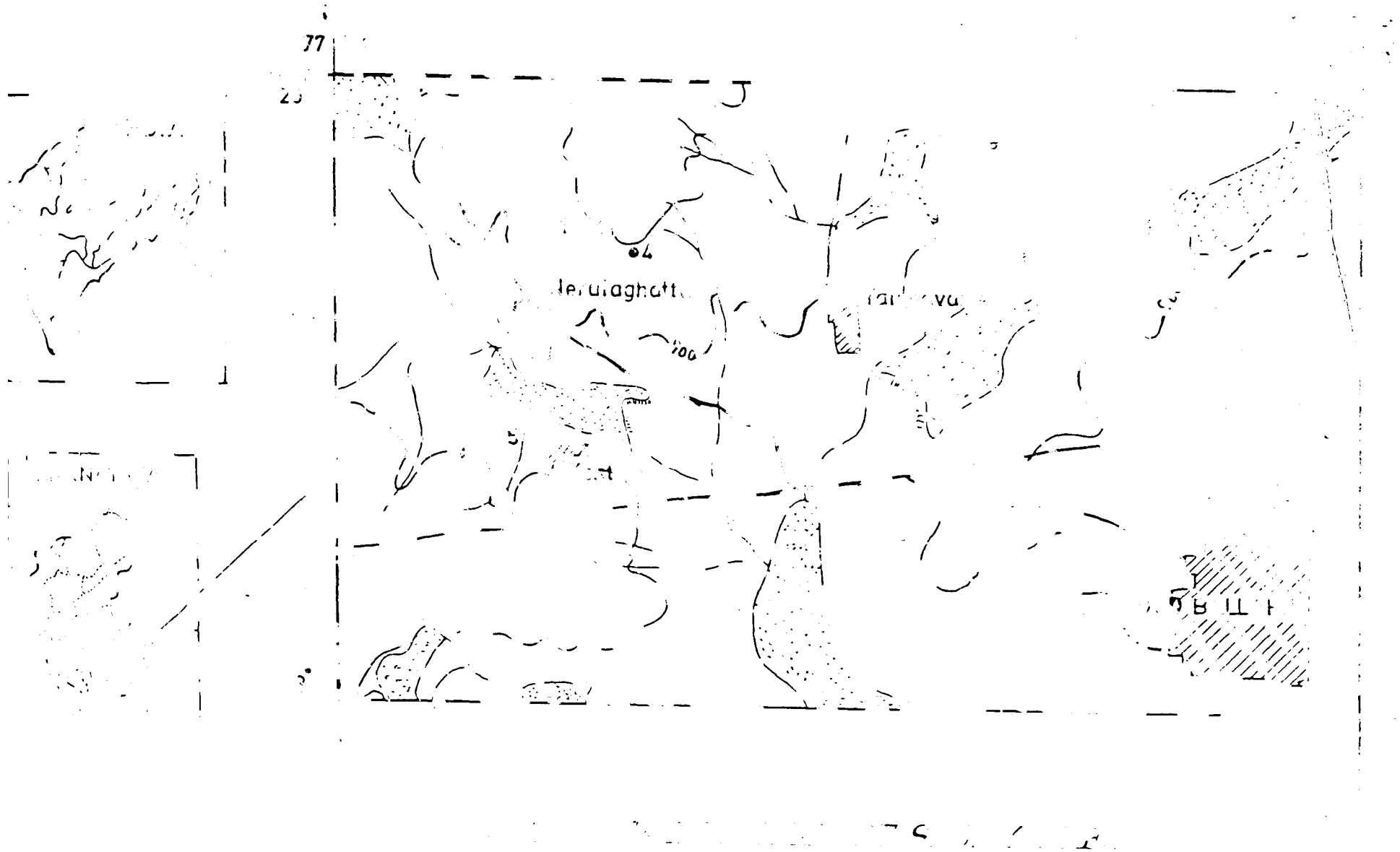
3.1 General description of the study area

Institute village Linkage Programme Area, Doddaballapura taluk, Bangalore rural district, present area of investigation, comprises two villages namely Neralaghatta and Kestur, spread over 3000 hectares, representing eastern dry zone of Karnataka State (Fig 1).

Neralaghatta village lies between 13°18'50" N latitudes and 77°29'0" E longitudes and covers an area of 1750 hectares. Kestur village which lies between 13°18'6" N latitude and 77°29'8" E longitudes and covers an area of 1250 hectares. The area receives annual average rainfall of 800-900 mm, spread over a period of six months with two peaks, one during the month of August another during the month of September. Soil temperature regime of the study area is Isohyperthermic (22°C to 30°C) and soil moisture regime is Ustic. The climatic data of study area is presented in Table 1 and graphically represented in Fig.3.

3.2 Geology

Karnataka forms part of peninsular shield, which is composed of geologically ancient rocks of diverse origin. Due to of varied climatic and physiographic conditions, these rocks have undergone various degrees of metamorphism and crushing resulting in the occurrence of different. The important geological formation of the study area belongs to Archean group. Granites and gneisses are two important rock types observed in the area.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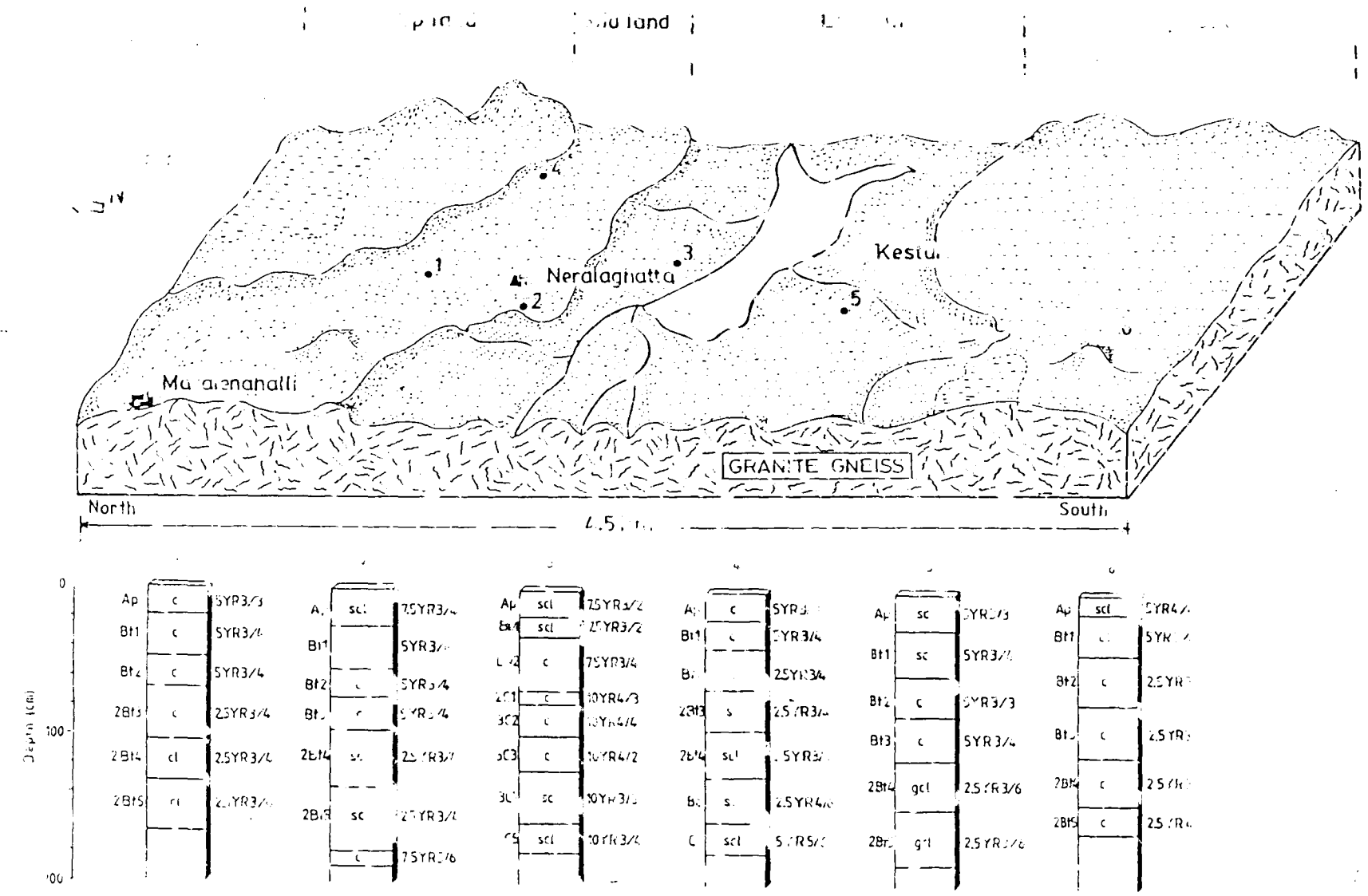


FIG. THEMATIC DIAGRAM OF SOIL PHYSIOGRAPHIC RELATIONSHIPS IN STUDY AREA

Table 1. Climatic data of study area

Area	January	February	March	April	May	June	July	August	September	October	November	December	Total
P	5.1	10.4	9.4	34	77.7	64.8	88.7	108.7	148.7	133.1	53.1	8.4	742.1
PE	117	130	166	158	157	127	116	114	109	105	98	103	1500
AE	12.1	11.78	9.65	34.03	77.7	64.8	88.7	108.7	109	105	85.96	34.7	742.1
°C	20.9	23.1	25.6	27.3	26.9	24.3	23.2	23.2	23.2	23.2	21.7	20.5	23.6

(Source: Doddaballapura Meteorological Station)

P = Precipitation
PE = Potentio evapotranspiration
AE = Actual evapotranspiration
°C = Temperature

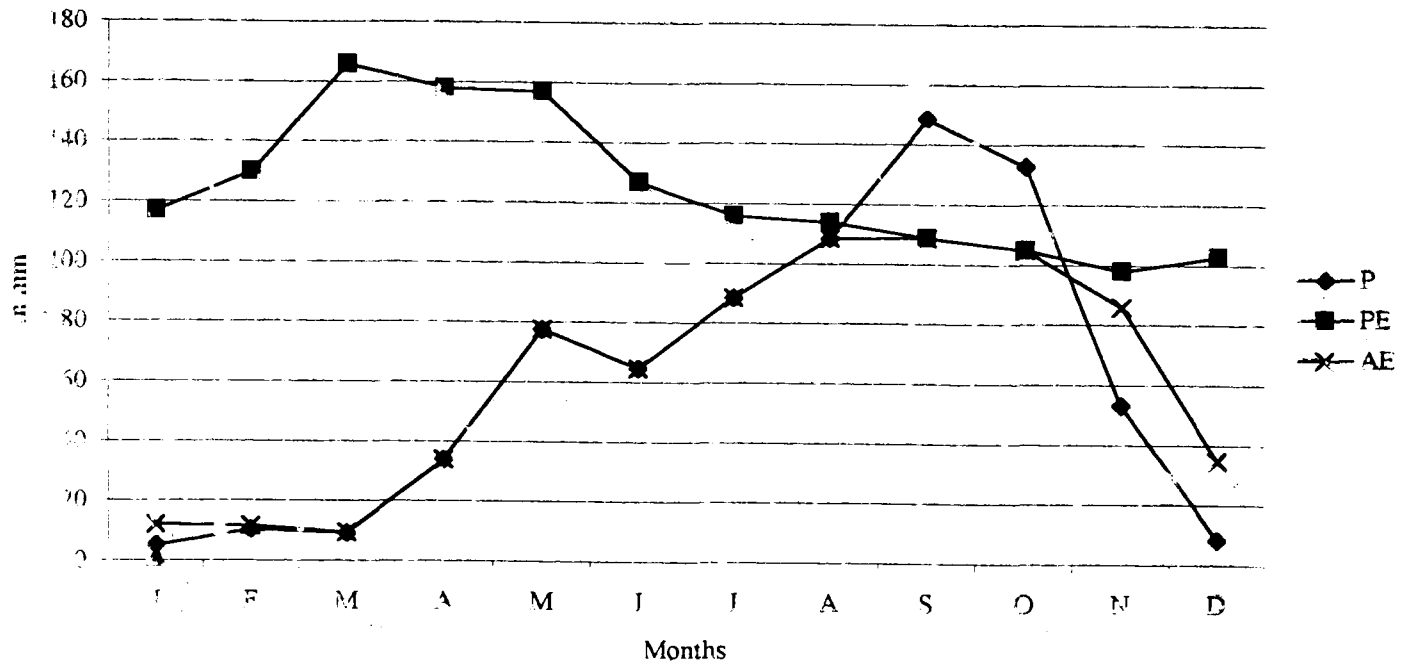


Fig 3: Climatic water balance diagram of Study Area

The area is gently sloping with 3-5 per cent slopes in the uplands, very gently sloping towards the low lands with 1-3 per cent slopes and nearly level with less than one per cent in the valley areas.

3.3 Present land use of study area

Since the area comes under rainfed region, present land use in Neralaghatta village is mainly ragi intercultivated with pigeon pea jowar, maize, horsegram, sesamum, wherever irrigation facility is there tomato, cabbage, chilli, paddy, sugarcane are grown in small patches as cash crops. In Kestur village present land use is ragi intercultivated with pigeonpea, pigeonpea as main crop, maize, jowar, sesamum under rainfed conditions and banana, cabbage, brinjal tomato, chilli under irrigated conditions.

3.4 Soil sampling

The major physiographic divisions occurring in the study area are uplands, midlands, lowlands and narrow valleys. In this gently slopping uplands, occupied major part of the landscape followed by uplands and lowlands. After traversing the entire IVLP area, two transacts representing these physiographic units were selected for the study. The soils were classified based on the soil taxanomy. Out of the six profiles studied three profiles classified as Typic Haplustalfs (profile nos.1, 4 and 5). Each profile as Kandic Paleustalfs (profile No.2), Typic Haplustepts (profile No.3) and Typic Paleustalfs (profile No.6).

Soil samples from six representative profiles covering the major land forms and soil types of Institute Village Linkage Programme Area, Doddaballapura taluk, Bangalore rural district were collected for the present study. Four pedons, samples collected from Neralaghatta tahasil out of which two pedons are representative of upland physiography (3.5% slope) and one each pedon from midland and low land physiography (1-3% and 0-1%, respectively). Two pedons, samples were collected from Kestur tahasil, out of which one pedon is representative of upland physiography (5-8% slope) and other pedon representing gently sloping (Slope 3-5%) physiography. These comprised of 40 samples from six selected profiles, which were collected horizon wise separately.

3.5 Preparation of soil samples

The representative soil samples collected from individual horizons of the profiles were air dried first. The sample was then thoroughly mixed, gently grounded with wooden mortar and pestle, taking care not to break primary particles. The sieved samples were thoroughly mixed again and preserved for further analysis.

3.6 Soil physical analysis

3.6.1 Particle size analysis

The particle size analysis was carried out by adopting the international pipette method using hydrogen peroxide as oxidizing agent for removing organic matter and sodium hexametaphosphate as dispersing agent (Piper, 1966).

3.6.2 Bulk density

The Bulk density was analysed by adopting core sampler method as outlined by Jackson (1973).

3.7 Soil chemical analysis

3.7.1 Soil reaction

The soil pH was determined in both soil : water suspension (1 : 2.5) and 1 N KCl (1 : 2.5)using pH meter (Jackson, 1973).

3.7.2 Electrical conductivity

The electrical conductivity values were determined from soil water suspension of 1 : 2.5 using Conductivity Bridge and values were expressed as dSm^{-1} at $25^{o}C$.

3.7.3 Organic carbon

The organic carbon was determined as per Walkly and Black wet digestion method (Jackson, 1973)

3.7.4 Cation exchange capacity

The cation exchange capacity of the soils were determined by saturating the exchange sites with ammonium ion (Jackson, 1973).

$$\%\ \text{CEC C.mol (p}^{+}\text{) kg}^{-1}\ \text{clay} = \frac{\text{CEC C.mol (p}^{+}\text{) kg}^{-1}\ \text{Soil}}{\text{Clay percentage}} \times 100$$

3.7.5 Exchangeable cations

The exchangeable cations were determined by leaching the soil with neutral normal ammonium acetate solution. Exchangeable calcium and magnesium in the leachate were determined by Atomic Absorption Spectrophotometer (Black, 1965). Exchangeable sodium and potassium were determined by flame photometer (Jackson, 1973).

3.7.5.1 Available nitrogen

Available nitrogen content of the soil was determined by following alkaline potassium permanganate method (Subbaiah and Asija, 1956).

3.7.5.2. Available phosphorus

In available phosphorus determination, extraction was done by using Bray`s-I extract and then subsequent estimation by ascorbic acid blue colour method (Jackson, 1973).

3.7.5.3 Available potassium

In the available potassium determination extraction was done by neutral normal ammonium acetate and subsequent estimation by flame photometer (Jackson, 1973).

3.7.5.4 Micronutrients

The micronutrients were determined by continuous shaking and leaching the soil with DTPA solution. The micronutrients thus extracted in the leachate were then determined by atomic absorption spectrophotometer (Lindsay and Norvell, 1969).

EXPERIMENTAL RESULTS

IV. EXPERIMENTAL RESULTS

Results of the analysis of morphological, physical and chemical characteristics of the six selected soil pedons are presented under the following headings.

4.1 Morphological properties of soil pedons.

4.2 Physical properties of soil pedons.

4.3 Chemical properties of soil pedons.

4.1 Morphological characteristics of soil pedons.

The detailed description of the morphological properties of the six selected soil profiles is presented in Table 2 their salient morphological characteristics are described under.

In pedon I, the surface horizon is upto 19 cm, with solum depth of 164 cm. The colour of soil was found to be reddish brown (2.5 YR 3/4 M). Texture of the profile varied from clay (first 4 horizons) to clay loam (last 2 horizons). Structure was found to be sub-angular blocky. The soil was found to be near neutral to acidic.

In pedon II, the solum depth recorded was 180 cm. The colour of the surface horizon was dark brown (7.5 YR 3/4 M) and the next two successive horizons Bt1 and Bt2 were dark reddish brown and reddish brown, respectively. In the subsequent horizons the colour was dark reddish brown. Texture of surface horizon was sandy clay loam where as subsurface horizons

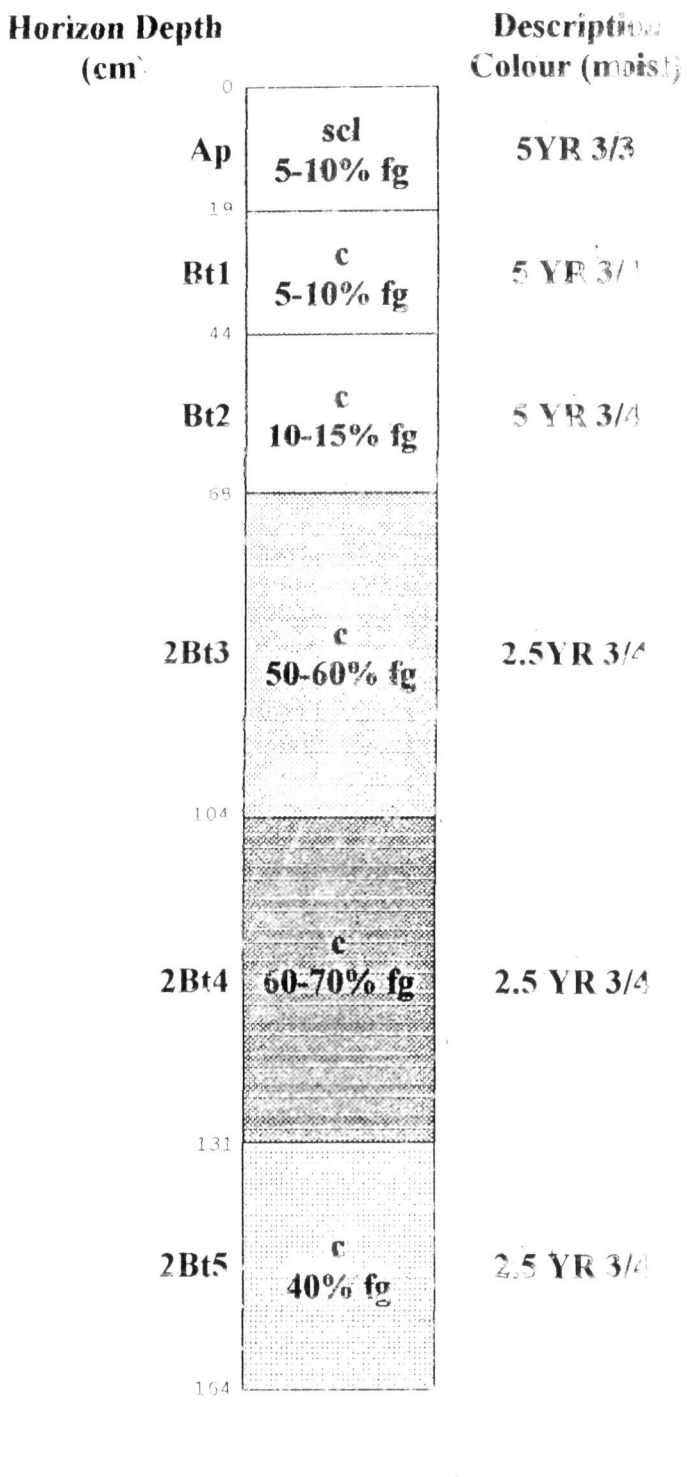


Fig.4. Profile diagram of Pedon 3

Description of Pedon-I

Location	Survey No.2, 500 m North from Neralaghatta village, 300 m west to the main road to the village Maralenahalli.
Physiography	Upland (slope 3-5%)
Parent material	Granite-Gneiss
Elevation	930 m MSL
Drainage	Well drained
Runoff	Medium
Land use	Ragi, jowar, maize
Natural vegetation	Jali, Jack, Neem, Pongamia
Climate	Tropical monsoon type with mean annual air temperature of 23.6°C. Mean annual rainfall 800-900 mm Soil moisture regime : Ustic Soil temperature regime – Isohyperthermic
Diagnostic horizons	0-19 cm ochric epipendon, 19-104 cm argillic horizon
Taxonomic classification	Fine, mixed, isohyperthermic family of Typic Haplustalfs

Typifying pedon – Pedon I Sandy clay loam - cultivated

Ap – 0 to19 cm	Dark reddish brown colour (5 YR 3/3) moist; fine, medium, weak subangular blocky structure; firm, slightly sticky; fine common roots; fine, medium and few common pores; about 5-10 per cent by volume of rock fragments; slightly acidic (pH 6.5) clear smooth boundary.
Bt1 – 19 to 44 cm	Dark reddish brown colour (5 YR 3/4) moist; medium, moderate subangular blocky structure; firm slightly plastic; fine and few roots; fine, medium and common pores; about 10 per cent by volume of rock fragments; slightly acidic (pH 6.80); clear smooth boundary.
Bt2 – 44 to 68 cm	Dark reddish brown colour (5 YR 3/4) moist; medium, moderate, subangular blocky structure; friable, very sticky and very plastic; fine and few roots; fine, medium and common pores; 10-15 per cent by volume of rock fragments; slightly acidic (pH 6.86); abrupt, smooth boundary.
2Bt3 – 68 to 104 cm	Dark reddish brown colour (2.5 YR 3/4) moist; medium, moderate, subangular blocky structure; friable, slightly plastic; fine few roots; fine medium and many pores; 50-60 per cent by volume of rock fragments; neutral soil reaction (pH 7.0); clear, smooth boundary.
2Bt4 – 104 to 131 cm	Dark reddish brown colour (2.5 YR 3/4) moist; medium, weak, subangular structure; friable slightly plastic; no roots; fine and many roots; neutral soil reaction (pH 7.10); abrupt smooth boundary.
2Bt5 – 131 to 164 cm	Dark reddish brown colour (2.5 YR 3/4) moist; medium, weak, subangular blocky structure; friable, slightly plastic; 40 per cent by volume of rock fragments.

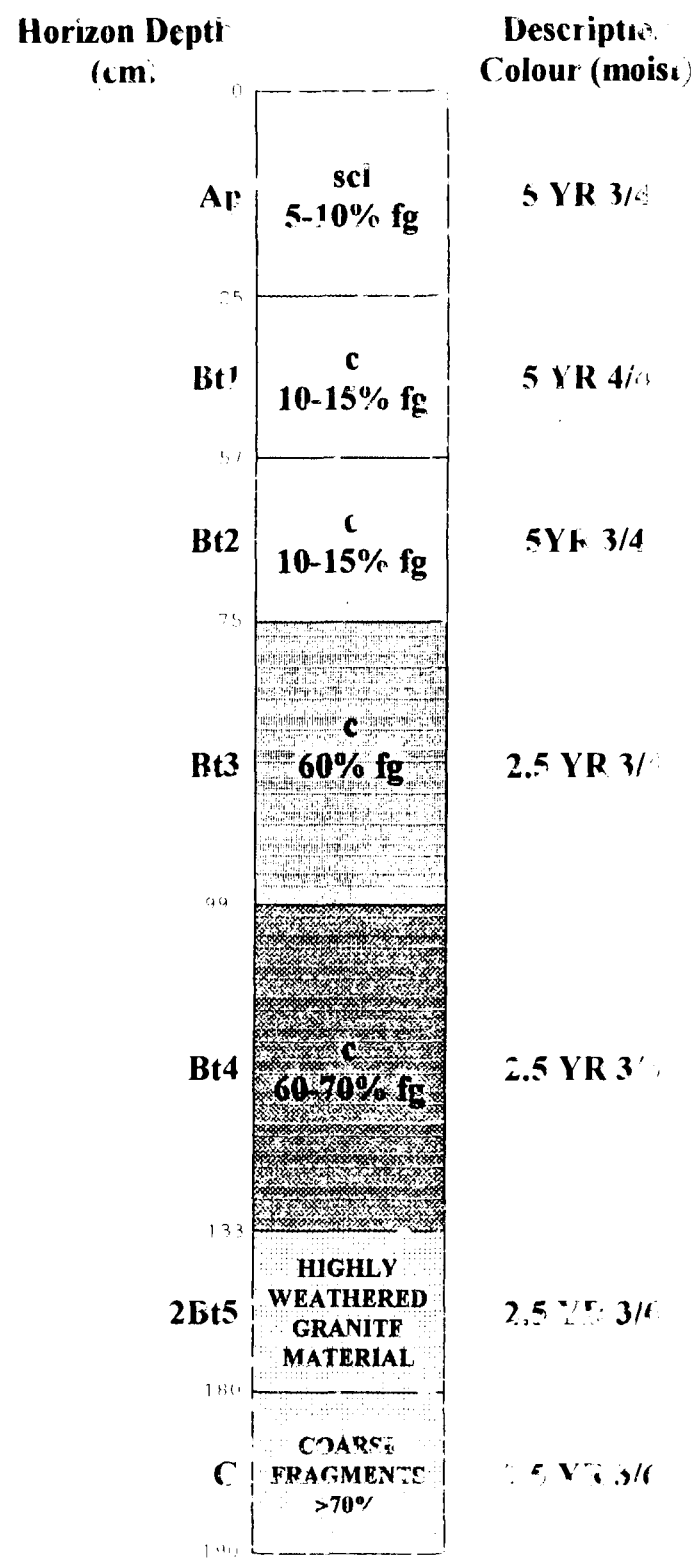


Fig.5. Profile diagram of Eden U

Description of Pedon-II

Location	Narappa's field, 600 m North from Neralaghatta village, 100 m West to road to village Maralenahalli.
Physiography	Midland, (Slope 1-3%)
Parent material	Granite-Gneiss
Elevation	925 m MSL
Drainage	Well drained
Runoff	Medium
Land use	Maize, Jowar, Cabbage, Chilli, Tomato
Natural vegetation	Bamboo, Pongamia
Climate	Tropical monsoon type with mean annual air temperature of 23.6°C. Mean annual rainfall 800-900 mm Soil moisture regime : Ustic Soil temperature regime – Isohyperthermic
Diagnostic horizons	0-25 cm ochric epipedon. 25-133 cm argillic horizon
Taxonomic classification	Fine, mixed, isohyperthermic family of Kandic Paleustalfs

Typifying pedon - Pedon II Sandy clay loam - cultivated

Ap – 0 to 25 cm	Dark brown colour (5YR 3/4) moist; sandy clay loam; fine, medium, weak subangular blocky structure; very friable, slightly plastic; fine and few roots; fine, medium and many pores; no gravels; neutral in soil reaction (pH 7.0); clear, smooth boundary.
Bt1 – 25 to 57 cm	Dark reddish brown (5 YR 3/4) colour moist; clay; medium, moderate, subangular blocky structure; friable and slightly plastic; fine, few roots; fine and medium, many pores; 5-10 per cent by volume of rock fragments; neutral soil reaction (pH 7.0); clear smooth boundary.
Bt2 – 57 to 75 cm	Reddish brown colour (5 YR 4/4) moist; clay; medium, moderate subangular blocky structure; friable and slightly plastic; fine, few roots; fine, medium and many pores; 10-15 per cent gravels by volume of rock fragments; slightly acidic (pH 6.5) in soil reaction; clear smooth boundary.
2Bt3 – 75 to 99 cm	Dark reddish brown colour (5 YR ¾) moist; clay, medium, moderate subangular blocky structure; friable, slightly plastic; no roots; fine medium and many pores; 10-15 per cent by volume of rock fragments; slightly acidic (pH 6.5) in soil reaction; abrupt, smooth boundary
2Bt4 – 99 to 133 cm	Dark reddish brown colour (2.5 YR ¾) moist; sandy clay; medium, weak, subangular blocky structure; friable, slightly plastic; no roots; fine, medium and many pores; 60 per cent by volume of rock fragments; neutral in reaction (pH 7.0); clear smooth boundary.
2Bt5 – 133 to 180 cm	Dark red colour (2.5 YR 3/6) moist; sandy clay; medium, weak, subangular blocky structure; friable slightly plastic; no roots; fine medium and many pores; 60-70 per cent by volume of rock fragments; slightly acidic (pH 6.5); clear, smooth boundary.
C – 180 to 190 cm	Highly weathered granite-gneiss parent material; strong brown colour (7.5 YR 3/6) moist; sandy clay; medium, weak, subangular blocky structure; friable, slightly plastic; no roots; fine, medium and many pores; more than 70 per cent by volume of rock fragments; slightly acidic (pH 6.5) abrupt, smooth boundary

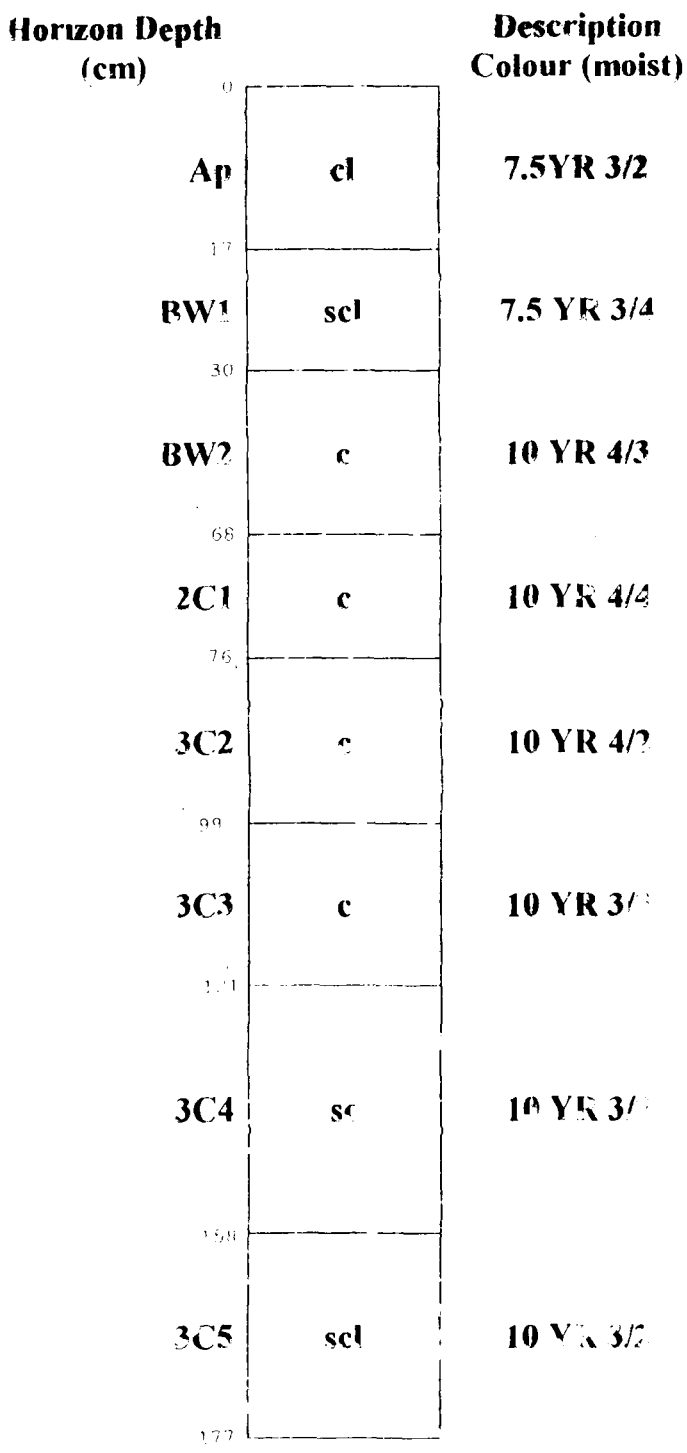


Fig.6. Profile diagram of Pedon 11

Description of Pedon-III

Location	Mr.Krishnappa's field, 400 m North from Neralaghatta village, 200 m west to the main road to the village Maralenahalli
Physiography	Low land (Slope 0-1%)
Parent material	Granite – Gneiss
Elevation	920 m MSL
Drainage	Moderately drained
Runoff	Very slow
Land use	Paddy, Sugarcane, Jowar, Ragi, Tomato
Natural vegetation	Acacia, Jack, shrubs, Mango, Pongamia
Climate	Tropical monsoon type with mean annual air temperature of 23.6°C. Mean annual rainfall 800-900 mm Soil moisture regime : Ustic Soil temperature regime – Isohyperthermic
Diagnostic horizons	0-17 cm ochric 17-68 cm cambic horizon
Taxonomic classification	Fine, mixed, Isohyperthermic family of Typic Haplustepts.

Typifying pedon - Pedon III Sandy clay loam - cultivated

Ap – 0 to 17 cm	Dark brown colour (7.5 YR 3/2) moist; sandy clay loam; medium, moderate, subangular blocky structure; friable, slightly sticky; fine and few roots; fine and common pores; no coarse fragments; soil reaction was neutral (pH 7.0); clear smooth boundary.
BW1 – 17 to 30 cm	Dark brown colour (7.5 YR 3/4) moist; sandy clay loam; medium, moderate, subangular blocky structure; friable, slightly plastic; fine and few roots; fine and common pores; no coarse fragments; neutral soil reaction (pH 7.0); clear smooth boundary.
BW2 – 30 to 68 cm	Dark brown colour (10 YR4/3) moist; clay; medium, strong, subangular blocky structure; friable slightly plastic; fine and few roots; very fine and common pores; no coarse fragments; slightly acidic (pH 6.5); clear, smooth boundary.
2C1 – 68 to 76 cm	Dark yellowish brown (10 YR 4/4) colour moist; clay; medium, moderate, subangular blocky structure; friable, slightly plastic; fine and few roots; fine and medium pores; no coarse fragments; slightly acidic (pH 6.5)
3C2 – 76 to 99 cm	Dark greyish brown (10 YR 4/2) colour moist; clay; massive, wedge shaped aggregates; very fine, very stick, very plastic; no roots; very fine and common pores; neutral soil reaction (pH 7.0)
3C3 – 99 to 121 cm	Dark brown colour (10 YR 3/3) moist; clay; massive, wedge shaped aggregates; very fine, very sticky, very plastic; no roots; very fine and common pores; no coarse fragments; neutral (pH 7.0) soil reaction.
3C4 – 121 to 158 cm	Very dark greyish brown colour (10 YR 3/2) moist; sandy clay; massive, wedge shaped aggregates; very fine, very sticky, very plastic; no roots; no coarse fragments; soil reaction was neutral (pH 7.0).
3C5 – 158 to 177 cm	Very dark greyish brown colour (10 YR 3/2) moist; sandy clay loam; massive, wedge shaped aggregates; very fine, very sticky, very plastic; no roots; very fine and common pores; slightly acidic (pH 6.5).

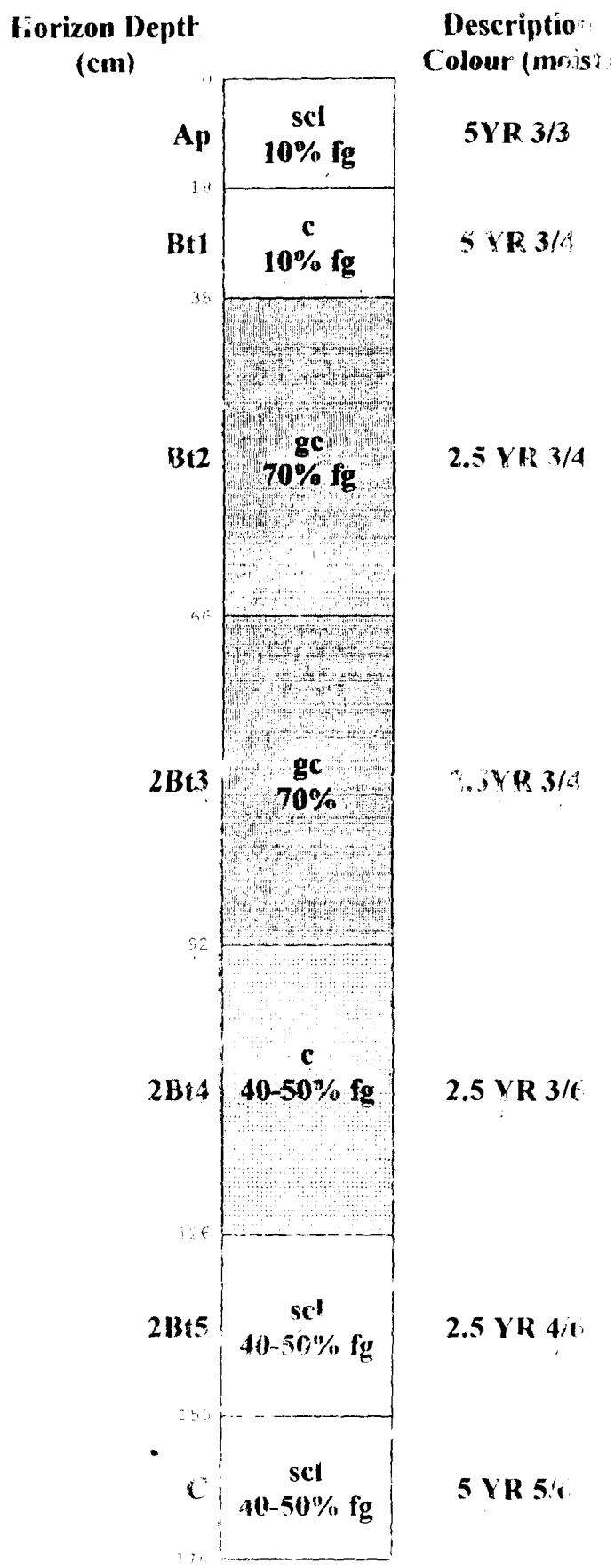


Fig. 7. Profile diagram of Pedon 15

Description of Pedon-IV

Location	Greentech Seeds Ltd., Farm, 600 m South to the village Neralaghatta, 200 m East to the main road which connects to Honnaghatta village road
Physiography	Upland (Slope 5-8%)
Parent material	Granite-Gneiss
Elevation	930 m MSL
Drainage	Well drained
Runoff	Rapid
Land use	Presently production of seeds of tomato, brinjal, maize
Natural vegetation	Tamarind, Jali, Pongamia
Climate	Tropical monsoon type with mean annual air temperature of 23.6°C. Mean annual rainfall 800-900 mm Soil moisture regime : Ustic Soil temperature regime – Isohyperthermic
Diagnostic horizons	0-18 cm ochric epipedon 18-26 cm argillic horizon
Taxonomic classification	Fine, mixed, isohyperthermic family of Typic Haplustalfs.

Typifying pedon - Pedon IV Sandy clay loam – cultivated

Ap – 0 to 18 cm	Dark reddish brown colour (5YR 3/3) moist; clay; fine, medium, moderate, subangular blocky structure; friable, slightly plastic; fine, medium and common roots; fine medium, many pores; 10 per cent by volume of rock fragments; neutral soil reaction (pH 7.0); clear smooth boundary.
Bt1 – 18 to 38 cm	Dark reddish brown colour (5 YR 3/4) moist; clay; medium, strong, subangular blocky structure; friable, very sticky, plastic; fine common roots; fine medium and many pores; 10 per cent by volume of rock fragments; slightly acidic (pH 6.5); clear, wavy boundary.
2Bt2 – 38 to 66 cm	Dark reddish brown colour (2.5 YR 3/4) moist; gravely clay; medium, moderate strong subangular blocky structure; very friable, very sticky and plastic; very fine, few roots; fine medium and many pores; 70 per cent by volume of rock fragments; slightly acidic; clear, wavy boundary.
2Bt3 – 66 to 92 cm	Dark reddish brown colour (2.5 YR 3/4) moist; gravely clay; medium, moderate, subangular blocky structure; very friable, very sticky and plastic; no roots; fine, medium and many pores; 70 per cent by volume of rock fragments; slightly acidic (pH 6.5); clear, wavy boundary.
2Bt4 – 92 to 126 cm	Dark red colour (2.5 YR 3/6) moist; gravely clay; medium, moderate, subangular blocky structure; very friable, slightly plastic; no roots; fine, medium and many pores; 40-50 per cent by volume of rock fragments; neutral soil reaction (pH 7.0); clear, wavy boundary.
2Bt5 – 126 to 155 cm	Red colour (2.5 YR 4/6) moist; sandy clay loam; medium, moderate, subangular blocky structure; very friable, slightly sticky and plastic; no roots; fine medium and many pores; 40-50 per cent by volume of rock fragments; neutral soil reaction; clear, wavy boundary.
C – 155 to 178 cm	Yellowish red colour (5 YR 5/6) moist; sandy clay loam; medium, moderate weak, subangular blocky; very friable, slightly sticky and plastic; no roots; fine medium many pores; 70 per cent by volume of rock fragments; slightly acidic (pH 6.5); clear wavy boundary

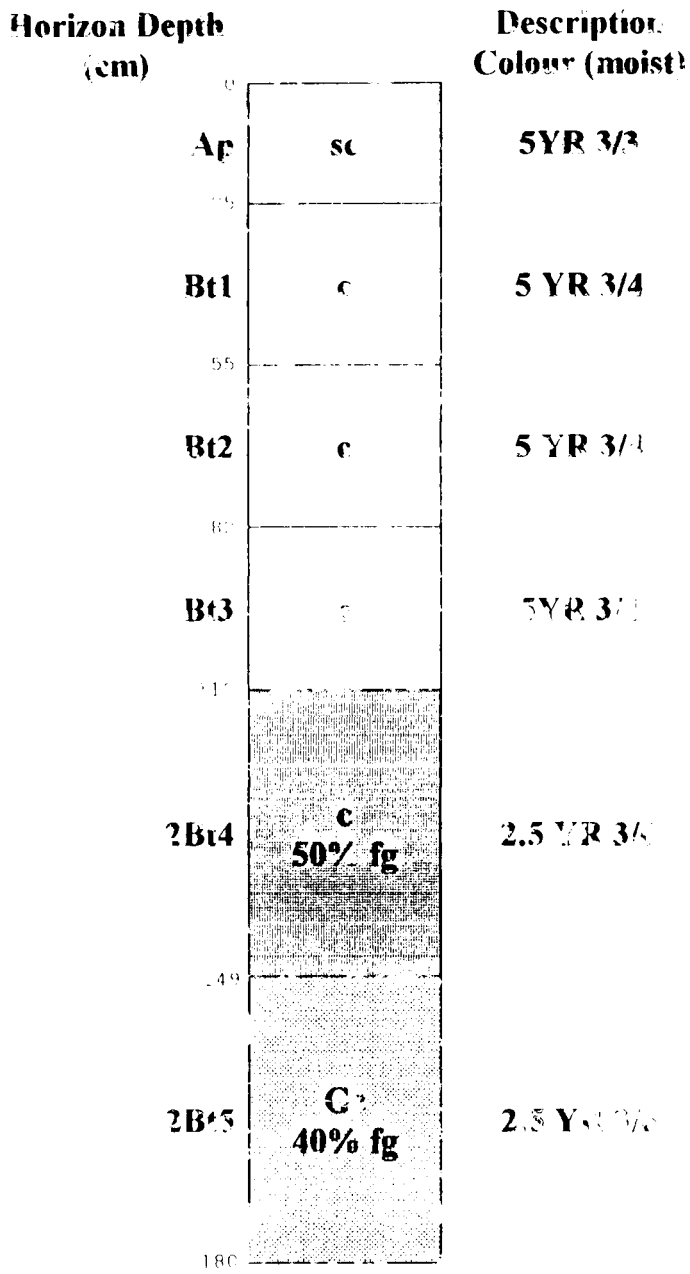


Fig.8 Profile diagram of [illegible]

Description of Pedon-V

Location	Chennakrishnappa's field Kestur village, 150 m East to the road connecting Anabe and Doddaballapura.
Physiography	Gently sloping (Slope 3-5%)
Parent material	Granite, Gneiss
Elevation	880 m MSL
Drainage	Well drained
Runoff	Medium
Land use	Ragi, Tomato, Banana, Cabbage
Natural vegetation	Tamarind, Pongamia, Banyan, Shrubs, Acacia, Jack
Climate	Tropical monsoon type with mean annual air temperature of 23.6°C. Mean annual rainfall 800-900 mm Soil moisture regime : Ustic Soil temperature regime – Isohyperthermic
Diagnostic horizons	0-25 cm ochric epipedon 25-149 cm argillic horizon
Taxonomic classification	Fine, mixed, Isohyperthermic family of Typic Haplustalfs.

Typifying pedon V – Pedon V Sandy clay - cultivated

Ap – 0 to 25 cm	Dark reddish brown colour (5 YR 3/3) moist; sandy clay; fine medium, moderate, subangular blocky structure; friable, slightly plastic; fine, few roots; fine, medium, many pores; no gravels; neutral soil reaction (pH 7.0); clear, smooth boundary.
Bt i – 25 to 55 cm	Dark reddish brown colour (5 YR ¾) moist; sandy clay; fine, medium, strong, subangular blocky structure; friable, slightly plastic; fine, few roots; fine, medium, pores; no gravels; neutral soil reaction; gradual, smooth boundary.
Bt2 – 55 to 82 cm	Dark reddish brown colour (5 YR 3/3) moist; clay; fine, medium, strong, subangular blocky structure; friable, very sticky and plastic; fine and few roots; fine, common pores; no gravels; neutral soil reaction (pH 7.0); gradual smooth boundary.
Bt3 – 82 to 112 cm	Dark reddish brown colour (5 YR 3/4) moist; clay; medium, strong, subangular blocky structure; friable, very sticky plastic; very fine, few roots; fine and many pores; no gravels; neutral soil reaction (pH 7.0); abrupt, smooth boundary.
2Bt4 – 112 to 149 cm	Dark red colour (2.5 YR 3/6) moist; gravely clay loam; fine and medium, moderate, subangular blocky structure; friable, slightly plastic; no roots; fine, medium, many pores; fine gravels 50 per cent by volume of rock fragments; slightly acidic (pH 6.5) in soil reaction; clear smooth boundary.
2Bt5 – 149 to 180 cm	Dark red colour (2.5 YR 3/6) moist; gravely clay loam; fine, medium moderate subangular blocky structure; friable, slightly plastic; no roots; fine, medium, many pores; fine gravels 40 per cent by volume of rock fragments; slightly acidic (pH 6.5); clear smooth boundary.

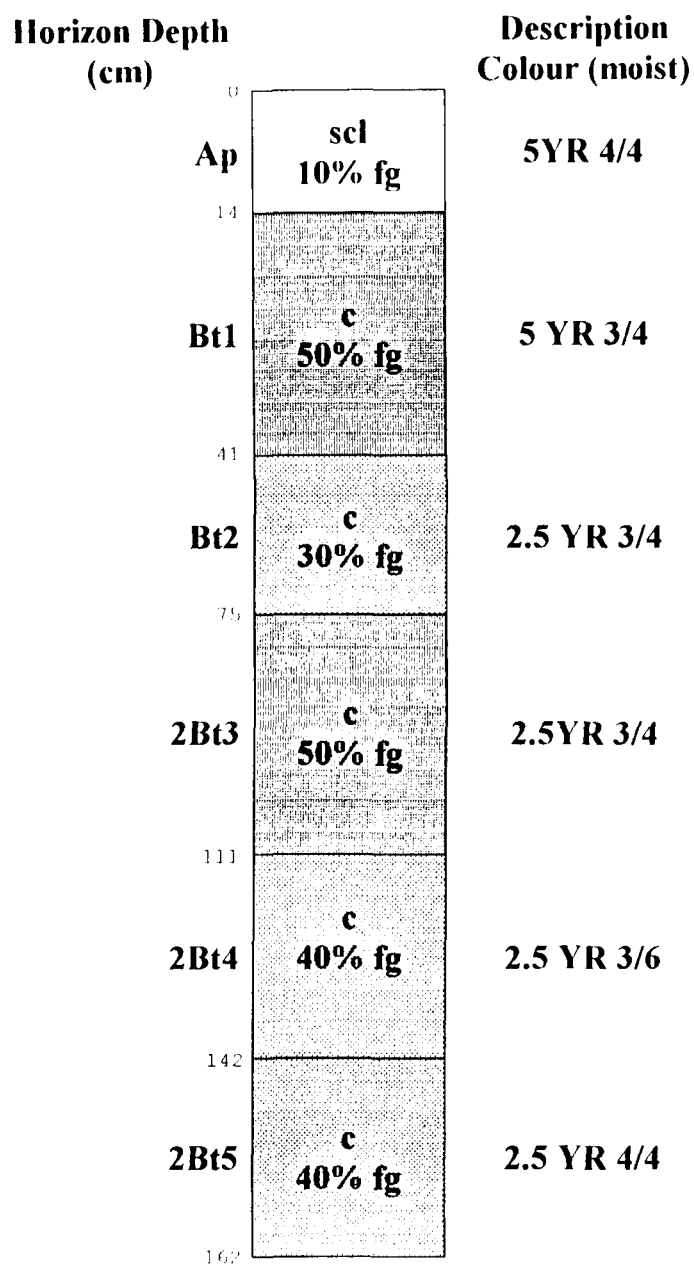


Fig.9. Profile diagram of Pedon VI

Description of Pedon-VI

Location	Narasappa's field (Doddaballapura to Sompura road) 1 km from Anabe cross (Belongs to Kestur village Mandal Panchayat)
Physiography	upland (Slope 3-8%)
Parent material	Granite Gneiss
Elevation	940 m MSL
Drainage	Well drained
Runoff	Rapid
Land use	Tur, Ragi, Cotton, Jowar, Horsegram
Natural vegetation	Eucalyptus, Acacia sps. Tamarind
Climate	Tropical monsoon type with mean annual air temperature of 23.6°C. Mean annual rainfall 800-900 mm Soil moisture regime : Ustic Soil temperature regime – Isohyperthermic
Diagnostic horizons	0-14 cm ochric epipedon 14-164 cm argillic horizon
Taxonomic classification	Fine, mixed, isohyperthermic family of Typic Paleustalfs.

Typifying pedon - Pedon VI Sandy clay loam - cultivated

Ap – 0 to 14 cm	Dark reddish brown colour (5 YR 4/4) moist; sandy clay loam; fine, moderate, subangular blocky structure; friable, slightly sticky, slightly plastic; fine common roots; fine, common pores; fine gravels 10 per cent by volume of rock fragments; slightly acidic (pH 6.5); gradual smooth boundary.
Bt1 – 14 to 41 cm	Dark reddish brown colour (5 YR 3/4) moist; clay loam; medium, strong, subangular blocky structures; friable, slightly plastic; very fine, common roots; fine, common pores; fine gravels 15 per cent by volume of rock fragments; neutral soil reaction (pH 7.0); gradual, smooth boundary.
2Bt2 – 41 to 75 cm	Dark reddish brown colour (2.5 YR 3/4) moist; clay; medium, strong, sub angular blocky; friable, slightly plastic; very fine, few roots; fine, common, pores; fine gravels 30 per cent by volume of rock fragments; neutral soil reaction (pH 7.0); abrupt, wavy boundary.
2Bt3 – 75 to 111 cm	Dark reddish brown colour (2.5 YR 3/4) moist; clay; medium, moderate, subangular blocky structure; friable slightly plastic; very fine, many roots; very fine common, many pores; fine gravels 50 per cent by volume of rock fragments; clear, wavy boundary.
2Bt4 – 111 to 142 cm	Dark red colour (2.5 YR 3/6) moist; medium, clay; moderate, subangular blocky structure; friable, very sticky, plastic; very fine, few roots; fine, common pores; coarse gravels 40 per cent by volume of rock fragments; slightly acidic in soil reaction (pH 6.5); clear smooth boundary.
2Bt5 – 142 to 162 cm	Reddish brown colour (2.5 YR 4/4) moist; clay; medium, moderate, subangular blocky structure; friable, very stick plastic; very fine, few roots; fine common pores; coarse gravels 40 per cent by volume of rock fragments; slightly acidic (pH 6.5); clear, smooth boundary.

had clay texture except in 2Bt4 and 2Bt5, where it was sandy clay. Presence of argillic horizon was noticed from 25 cm to 180 cm. Both surface and subsurface horizons exhibited sub-angular blocky structure. Soils were slightly acidic to neutral in reaction.

The solum depth in pedon III was 68 cm. The colour of the surface as well as subsurface horizon was dark brown, where as in next successive horizons it ranged from brown to very dark grayish brown. Surface and subsurface horizons showed sandy clay loam texture. Sub-angular blocky structure was observed in the surface horizon and continued upto 2C1 horizon. From 2C2 horizon onwards massive wedge shaped aggregates were present upto 3C5 horizon. Soils were slightly acidic to near neutral in reaction. The physiography observed was low land (0-1% slope).

In pedon IV solum depth was 155 cm, colour of the soils ranged from dark reddish brown (surface horizon) to red colour (subsurface horizons). Clay texture is noticed upto 66 cm from surface, there onwards texture changed to gravelly clay. Structure of soil was found to be sub-angular blocky. Soils were slightly acidic to neutral in reaction.

In pedon V, solum depth was 180 cm, colour of soils varied between dark reddish brown (Ap horizon to Bt3 horizon) and red colour (Bt4 and 2Bt5 horizons). Structure of soil was found to be sub-angular blocky from surface horizon to throughout the depth. Texture of soil was sandy clay in first two horizons (Ap and Bt1 horizons), followed by clay in the lower horizons.

Table 2. Morphological characteristics of profiles

Profile No.	Physiography	Slope (%)	Horizon	Depth (cm)	Colour (Moist)	Structure	Consi stency	Boun dary	Porosity	Roots	Cutans	Coarse fragments (% by vol.)
01	Upland	3-5	Ap	0-19	5YR 3/3	fm1Sbk	fiss	c s	fm fc	fc	-	fg 5-10
			Bt1	19-44	5YR 3/4	m2Sbk	fisp	c s	fm c	ff	tk,c	f-10
			Bt2	44-68	5YR 3/4	m2Sbk	frsp	a s	fm c	ff	tk,c	f 20-15
			2Bt3	68-104	2.5YR 3/4	m1Sbk	frsp	c s	fm m	vff	mtk,p	fg 50-60
			2Bt4	104-131	2.5YR 3/4	m1Sbk	frsp	a s	f m	No	mtk,p	fg 60-70
			2Bt5	131-164	2.5YR 3/4	m1Sbk	frsp	-	f m	No	mtk,c	fg >70
02	Midland	1-3	Ap	0-25	5 YR 3/4	fm1 Sbk	mfr sp	c s	fm m	ff	T tk c	fg < 5
			Bt1	25-57	5 YR 3/4	m2 Sbk	fr sp	c s	fm m	ff	T tk c	fg 5-10
			Bt2	57-75	5 YR 4/4	m2 Sbk	fr sp	c s	fm m	ff	T tk c	fg 10-15
			Bt3	75-99	5 YR 3/4	m2 Sbk	fr sp	a s	fm m	-	T tk c	fg 10-15
			2Bt4	99-133	2.5 YR 3/4	m1 Sbk	fr sp	c s	fm m	-	T tk c	fg 60
			2Bt5	133-180	2.5 YR 3/6	m1 Sbk	fr sp	c s	fm m	-	T tk c	fg 60-70
			C	180-190	7.5 YR 3/6	-	fr sp	-	fm m	-	T tk c	fg >70

(According to soil survey staff, 1975)

*Structure: f – fine; m – medium; 1 – weak; 2 – moderate; sbk – subangular blocky.
Cutans: T – argillian; tk – thick; c – continuous; p – patchy; pc – patchy continuous; mtk – moderately thick
Consistency: fl – firm; fs – friable; ss – slightly sticky; sp – slightly plastic.
Boundary: c – clear; a – abrupt; s – smooth.
Porosity: f – fine; vf – very fine; m – medium; c – coarse; c – common.
Coarse fragments: fg – fine gravel (<2.5 cm); Cg – coarse gravel (2.5-7.5 cm); c – coarse.

Table 2 (contd...)

Profile No.	Physiography	Slope (%)	Horizon	Depth (cm)	Colour (Moist)	Structure	Consistency	Boundary	Porosity	Roots	Cutans	Coarse fragments (% by vol.)
03	Lowland	0-1	Ap	0-17	7.5 YR 3/2	m1Sbk	fr ss	c s	fc	ff	-	-
			BW1	17-30	7.5YR 3/4	m2Sbk	fr sp	c s	fc	ff	-	-
			BW2	30-68	10YR 4/3	m2Sbk	fr sp	c s	vfc	ff	-	-
			2C1	68-76	10YR 4/4	m2Sbk	fr sp	c s	fm	ff	-	-
			3C2	76-99	10YR 4/2	massive	vfr vs vp	c s	vfc	-	-	-
			3C3	99-121	10YR 3/3	wedge	vfi vs vp	c s	vfc	-	-	-
			3C4	121-158	10YR 3/2	shaped	vfi vs vp	c s	vfc	-	-	-
			3C5	158-177	10YR 3/2	aggregates	vfi vs vp	c s	vfc	-	-	-
04	Upland	3-8	Ap	0-18	5YR 3/3	fm2Sbk	fr sp	c s	fm m	fmc	T tk c	fg<10
			Bt1	18-38	5YR 3/4	m3Sbk	fr vsp	c w	fm m	fc	T tk c	fg 10-70
			Bt2	38-66	2.5YR 3/4	m2Sbk	vfr vsp	-	fm m	vf f	T tk pc	fg 70
			2Bt3	66-92	2.5YR 3/4	m2Sbk	vfr vsp	-	fm m	-	T tk pc	fg 70
			2Bt4	92-126	2.5YR 3/6	m2Sbk	vfr sp	-	fm m	-	-	fg 40-50
			2Bt5	126-155	2.5YR 4/6	m2Sbk	vfr ssp	-	fm m	-	-	-
			C	155-178	5YR 5/6	m2Sbk	-	-	fm m	-	-	-

(According to soil survey staff, 1975)

*Structure: f – fine; m – medium; 1 – weak; 2 – moderate, sbk – subangular blocky.
Cutans: T – argillian; tk – thick; c – continuous; p – patchy; pc – patchy continuous; mtk – moderately thick
Consistency: fl – firm; fs – friable; ss – slightly sticky; sp – slightly plastic.
Boundary: c – clear; a – abrupt; s – smooth.
Porosity: f – fine; vf – very fine; m – medium; c – coarse; c – common.
Coarse fragments: fg – fine gravel (<2.5 cm); Cg – coarse gravel (2.5-7.5 cm); c – coarse.

Table 2 (contd.....)

Profile No.	Physiography	Slope (%)	Horizon	Depth (cm)	Colour (Moist)	Structure	Consi stency	Boun dary	Porosity	Roots	Cutans	Coarse fragments (% by vol.)
05		3-5	Ap	0-25	5YR 3/3	fm2sbk	fr sp	cs	fm m	ff	-	-
			Bt1	25-55	5YR 3/4	fm3sbk	fr sp	gs	f m	ff	tk c	-
			Bt2	55-82	5YR 3/3	fm3sbk	fr vsp	gs	f m	ff	tk c	-
			Bt3	82-112	5YR 3/4	m3sbk	fr sp	as	f m	vf f	tk c	-
			Bt4	112-149	2.5YR 3/6	fm2sbk	fr sp	cs	fm m	-	tk c	fg 50
			Bt5	149-180	2.5YR 3/6	fm2sbk	fr sp	-	fm m	-	tk c	fg 40
06	Upland	3-8	Ap	0-14	5YR 4/4	f2 sbk	fr ss ps	gs	f c	fc	t tn p	fg cg 10
			Bt1	14-41	5YR 3/4	m3 sbk	fr sp	gs	f c	vfc	t tn p	fg cg 50
			Bt2	41-75	2.5YR 3/4	m2 m	fr sp	aw	f c	vff	t tk c	fg cg 30
			2Bt3	75-111	2.5YR 3/4	m2 sbk	fr sp	cw	vfc mf	vfm	t tk c	fg cg 50
			2Bt4	111-142	2.5YR 3/6	m2 sbk	fr sp	cs	m f	vfc	t tk c	fg cg 40
			2Bt5	142-162	2.5YR 4/4	m2 sbk	fr vsp	cs	f c	vff	t tk c	fg cg 40

(According to soil survey staff, 1975)

*Structure: f – fine; m – medium; 1 – weak; 2 – moderate; sbk – subangular blocky.
Cutans: T – argillian; tk – thick; c – continuous; p – patchy; pc – patchy continuous; mtk – moderately thick
Consistency: fl – firm; fs – friable; ss – slightly sticky; sp – slightly plastic.
Boundary: c – clear; a – abrupt; s – smooth.
Porosity: f – fine; vf – very fine; m – medium; c – coarse; c – common.
Coarse fragments: fg – fine gravel (<2.5 cm); Cg – coarse gravel (2.5-7.5 cm); c – coarse.

Structure was found to be sub-angular blocky throughout the depth from surface horizon. Soils were neutral to slightly acidic in reaction.

In pedon VI, solum depth extended upto 162 cm, colour of soils ranged from reddish brown (surface horizon) to dark reddish brown (sub surface horizons). Texture of soil was noted to be sandy clay loam in the surface horizon whereas in sub-surface horizons it was clay throughout the depth of the profile. Soil structure was sub-angular blocky in all the horizons except in the last horizon. The soils are slightly acidic to near neutral in reaction.

4.2 Physical characteristics of soil profiles

4.2.1 Particle size distribution

The results of mechanical analysis of the six selected pedons are presented in Table 3.

4.2.1.1 Clay

Clay content of the pedon I was 37.0 per cent in surface horizon. It increased with depth from 48.2 to 52.2 per cent till Bt3 horizon, however later on showed irregular pattern of distribution. In pedon II the clay content varied from 32.5 to 52.1 per cent, respectively. This showed an increasing trend with the depth till 2Bt3 horizon, afterwards showed decreasing trend with depth. In pedon III, 28.9 and 23.6 per cent of clay was present in Ap and BW1 horizons, respectively. Later on it increased with increase in depth (47.4, 50 and 52.1%) and then decreased with further increase in depth. Clay content of

pedon IV showed increasing trend with depth right from surface horizon and extended upto 2Bt2 horizon. However, it decreased with depth of profile regularly. In pedon V clay content ranged between 37.6 to 49.6 per cent and showed increasing trend with depth extending upto Bt2 horizon afterwards decreasing trend with increase in depth was noticed. Clay content of pedon VI showed increasing trends with depth from surface horizon extending upto Bt2 horizon. Later on irregular distribution was observed in 2Bt3, 2Bt4 and 2Bt5 (51.6, 55.4 and 44.0%) horizons.

4.2.1.2 Silt

The silt content of pedon I varied between 7.6 to 23.8 per cent. In the surface horizon it was 7.6 per cent, highest silt content was present in 2Bt4 horizon i.e., 23.8 per cent.

In pedon II surface horizon contained 5.6 per cent of silt afterwards it decreased with depth till 2Bt4 horizon then increased in the last two horizons, 8.5 per cent in 2Bt5 and 12.3 per cent C horizon.

In pedon III silt content showed irregular pattern of distribution. It was 19.2 per cent in surface horizon followed by 21.0, 17.2, 25.4, 21.4, 29.5, 24.2 and 8.7 per cent of silt in BW1, BW2, 2C1, 3C2, 3C3, 3C4 and 3C5 horizons, respectively.

Silt content of pedon IV also showed irregular pattern of distribution as highest silt content was present in surface horizon and lowest in 2Bt4 horizon i.e., 10.5 per cent.

In pedon V silt content showed increasing trend with the depth from surface horizon and varied between 8.9 in the surface horizon and 39.0 per cent in the 2Bt5 horizon.

In pedon VI silt content did not draw any regular increasing or decreasing trend with depth. The highest silt content was present in Bt1 horizon (25.8%) and lowest was in 2Bt4 horizon (6.4%).

4.2.1.3 Sand

The sand content of pedon I showed decreasing trend with depth of profile as highest sand was present in Ap horizon and lowest content was present in Bt2 horizon i.e., 31.8 per cent.

In pedon II, the sand content showed decreasing trend with depth of profile, it varied between 61.9 per cent in Ap horizon and lowest amount of 40.8 per cent in Bt1 horizon.

In pedon III, the sand content was highest in BW1 horizon (55.4%) whereas, it was lowest in 2C1 horizon 24.6%). The sand content decreased from subsurface horizon (BW1 horizon) to 3C2 horizon.

Table 3. Bulk density and particle size distribution of pedons

Profile No	Horizon	Depth Cm	Bulk density (g/cc)	Particle size distribution (%)			
				Sand (2-0.5 mm)	Silt (0.05-0.002 mm)	Clay (<0.002 mm)	Textural class
01	Ap	0-19	1.57	55.4	7.6	37.0	sc
	Bt1	19-44		41.4	11.9	48.2	c
	Bt2	44-68		31.8	18.4	49.8	c
	Bt3	68-104		36.6	11.2	52.2	c
	2Bt4	104-131		32.1	23.8	38.1	cl
	2Bt5	131-164		44.6	16.5	38.9	cl
02	Ap	0-25	1.59	61.9	5.6	32.5	scl
	Bt1	25-57		40.8	11.1	48.1	c
	Bt2	57-75		41.0	8.4	50.6	c
	Bt3	75-99		43.2	4.7	52.1	c
	2Bt4	99-133		48.9	4.2	46.9	sc
	2Bt5	133-180		51.8	8.5	39.7	sc
	C	180-190		48.8	12.3	38.9	sc

Table 3 (contd.....)

Profile No	Horizon	Depth Cm	Bulk density (g/cc)	Particle size distribution (%)			
				Sand (2-0.5 mm)	Silt (0.05-0.002 mm)	Clay (<0.002 mm)	Textural class
03	Ap	0-17	1.58	51.2	19.2	29.6	scl
	BW1	17-30		55.4	21.0	23.6	scl
	BW2	30-68		35.6	17.2	47.2	c
	2C1	68-76		24.6	25.4	50.0	c
	3C2	76-99		26.5	21.4	52.1	c
	3C3	99-121		30.2	29.5	40.3	c
	3C4	121-158		37.1	24.2	38.7	sc
	3C5	158-177		58.8	8.7	32.5	scl
04	Ap	0-18	1.57	23.1	34.1	42.8	c
	Bt1	18-38		35.0	21.3	43.7	c
	2Bt2	38-66		25.0	22.2	52.8	gc
	2Bt3	66-92		42.5	19.3	38.2	gc
	2Bt4	92-126		55.5	11.6	32.9	gc
	Bc	126-155		58.0	10.5	31.5	scl
	C	155-178		59.2	10.7	30.1	scl

Table 3 (contd.....)

Profile No	Horizon	Depth cm	Bulk density (g/cc)	Particle size distribution (%)			
				Sand (2-0.5 mm)	Silt (0.05-0.002 mm)	Clay (<0.002 mm)	Textural class
05	Ap	0-25	1.51	53.5	8.9	37.6	sc
	Bt1	25-55		46.0	10.1	43.9	sc
	Bt2	55-82		29.3	21.1	49.6	c
	Bt3	82-112		29.1	22.4	48.5	c
	2Bt4	112-149		33.0	28.6	38.4	gcl
	2Bt5	149-180		23.4	39.0	37.6	gcl
06	Ap	0-14	1.49	56.7	14.3	29.0	scl
	Bt1	14-41		36.6	25.8	37.6	cl
	Bt2	41-75		27.3	19.1	53.6	c
	2Bt3	75-111		31.7	16.7	51.6	sc
	2Bt4	111-142		38.2	6.4	55.4	c
	2Bt5	142-162		40.0	16.0	44.0	c

In pedon IV sand content was 23.1 per cent in surface horizon, respectively. Increasing trend was observed from 2Bt2 horizon to C horizon i.e., 25.0 to 59.3 per cent, respectively.

The sand content of pedon V has decreased from 53.5 per cent in surface horizon to 23.4 per cent in subsurface 2Bt5 horizon.

In pedon VI, highest sand content was present in Ap horizon i.e., 56.7 per cent from where decreasing trend was observed upto 2Bt3 horizon.

4.3 Chemical characteristics of soil profiles

The data related to soil reaction (pH), electrical conductivity and organic carbon are presented in Table 4.

4.3.1 Soil reaction (pH)

pH of the soil estimated in 1 : 2.5 soil : water extract and 1 N KCl was noted to be between 5.4 to 7.1 in pedon I and showed an increasing trend with depth.

In pedon II, surface horizon showed a value of 7.1 where as Bt3 horizon showed 6.1 pH. Irregular pattern of pH was noticed both in soil : water extract and in 1 N KCl.

In pedon III, pH of soil ranged between 7.0 to 6.8 (1 : 2.5 soil : water extract) in 3C3 and 2 Cl horizons, respectively. Surface horizon showed pH of 6.9 whereas lower most 3C5 horizon showed pH of 7.0 in 1 : 25 soil : water extract.

Irregular pattern of pH was observed in pedon IV where surface horizon showed pH of 7.1 and subsurface, Bt1 horizon showed 6.9 pH. Later, it increased slightly to 7.0 (2Bt2 horizon) and remained constant in 2Bt3 and 2Bt4 i.e., 6.8 and 6.8, respectively.

Decreasing trend of pH was noticed in pedon V from surface horizon to throughout the depth. Lower most horizon (2Bt5) showed a value of 6.2 compared with 7.1 in Ap horizon.

In pedon VI first three horizons i.e., Ap, Bt1 and Bt2 showed increasing trend with depth (7.0, 7.0, 6.8, respectively) where as second three horizons i.e., 2Bt3, 2Bt4, and 2 Bt5 showed decreasing trend (6.8, 6.7, 6.6, respectively).

4.3.2 Electrical conductivity

The electrical conductivity of the soils estimated in 1 : 2.5 soil : water extract was found to be very low in all the pedons. In pedon I electrical conductivity value 0.02 dSm^{-1} and 0.04 dSm^{-1} at 25°C. In pedon II electrical conductivity values ranged from 0.02dSm^{-1} and 0.05 dSm^{-1} was the range of electrical conductivity values of pedon II. In pedon III electrical conductivity values ranged from 0.02 dSm^{-1} to 0.08 dSm^{-1}. The pedon IV electrical conductivity ranged from 0.04 dSm^{-1} to 0.07 dSm^{-1}. In pedon V electrical conductivity remained constant from subsurface Bt1 horizon throughout the depth with the exception of lower most 2Bt5 horizon where it was 0.06 dSm^{-1}.

The electrical conductivity ranged between 0.01 dSm^{-1} and 0.04 dSm^{-1} in pedon VI.

4.3.3 Organic carbon

The organic carbon content of the soils estimated were observed to be ranged from low to high in all the pedons (Table 4).

Ap horizon of pedon I contained higher percentage (0.91%)of organic carbon, which decreased regularly with depth.

In pedon II organic carbon content varied from 0.82 to 0.20 per cent between the surface and lower most (C) horizons. In pedon III decreasing trend of organic carbon content was observed right from the surface horizon (0.58%) to the 3C5 horizon (0.04%).

In pedon IV, subsurface Bt1 horizon contained higher percent (0.76%) of organic carbon which later on decreased with increasing depth of the soil surface horizon contained 0.54 per cent of organic carbon.

In pedon V, Surface and subsurface Bt1 horizons showed 0.62 and 0.55 per cent of organic carbon. Later it decreased regularly with increase in depth of profile.

In pedon VI, organic carbon content showed decreasing trend from subsurface Bt2 horizon extending upto the lower most 2Bt5 horizon. Surface horizon was having higher amount of organic carbon (0.70%).

Table 4. Chemical characteristics of the pedons

Profile No.	Horizon	Depth (cm)	pH		EC (dSm^{-1})	OC (%)
			1 : 2.5 H_2O	1 : 2.5 1 N KCl	1 : 2.5	
01	Ap	0-19	6.90	5.60	0.04	0.91
	Bt1	19-44	6.30	5.40	0.02	0.63
	Bt2	44-68	6.80	5.60	0.02	0.51
	Bt3	68-104	7.00	5.90	0.04	0.47
	2Bt4	104-131	7.10	6.10	0.04	0.36
	2Bt5	131-164	7.10	6.10	0.03	0.32
02	Ap	0-25	7.10	6.10	0.05	0.82
	Bt1	25-57	7.01	6.20	0.03	0.37
	Bt2	57-75	6.25	6.08	0.04	0.45
	Bt3	75-99	6.10	6.01	0.04	0.49
	2Bt4	99-133	7.00	6.06	0.04	0.24
	2Bt5	133-180	6.90	0.06	0.02	0.21
	C	180-190	6.87	6.05	0.04	0.20
03	Ap	0-17	6.91	6.18	0.07	0.58
	BW1	17-30	6.87	6.26	0.04	0.53
	BW2	30-68	6.90	6.50	0.02	0.50
	2C1	68-76	6.80	6.40	0.04	0.41
	3C2	76-99	7.00	6.60	0.07	0.38
	3C3	99-121	7.01	6.18	0.08	0.21
	3C4	121-158	6.97	6.15	0.04	0.08
	3C5	158-178	7.00	6.80	0.06	0.04
04	Ap	0-18	7.10	6.67	0.07	0.54
	Bt1	18-38	6.90	6.69	0.04	0.76
	2Bt2	38-66	7.01	6.80	0.06	0.51
	2Bt3	66-92	6.80	6.18	0.07	0.48
	2Bt4	92-12	6.80	6.18	0.07	0.08
	BC	126-155	7.01	6.19	0.07	0.04
	C	155-178	6.90	6.46	0.06	0.04
05	Ap	0-25	7.01	6.80	0.06	0.62
	Bt1	25-55	6.90	6.50	0.07	0.55
	Bt2	55-82	6.80	6.40	0.07	0.48
	Bt3	82-112	6.90	6.46	0.07	0.44
	2Bt4	112-149	6.50	6.90	0.07	0.44
	2Bt5	149-180	6.20	5.20	0.06	0.37
06	Ap	0-14	6.98	5.96	0.04	0.70
	Bt1	14-41	7.01	6.20	0.02	0.54
	Bt2	41-75	6.99	6.10	0.02	0.66
	2Bt3	75-111	6.80	5.60	0.02	0.59
	2Bt4	111-142	6.66	5.43	0.01	0.47
	2Bt5	142-162	6.59	5.53	0.04	0.39

4.3.4 Cation exchange capacity

The data related to cation exchange capacity, exchangeable calcium, magnesium, potassium and sodium and per cent base saturation are presented in Table 5. The cation exchange capacity of the soil was estimated using NH_4OAc extract.

In pedon I, it was found to be 8.35 c.mol (p^+) kg^{-1} and 19.11c.mol (p^+) kg^{-1} in Ap and Bt2 horizons, respectively which indicated an increasing trend with depth. However decreasing trend with the depth was observed in last three successive horizons.

In pedon II highest cation exchange capacity value was noted in Bt1 horizon (12.91 c.mol (p^+) kg^{-1}) and lowest value was noted in 2Bt5 horizon (7.09 c.mol (p^+) kg^{-1}). No significant or regular increasing or decreasing trend was observed.

In pedon III, lowest cation exchange capacity was noticed in the last horizon (8.08 c.mol (p^+) kg^{-1}). Irregular trend with the depth in cation exchange capacity was observed in this profile upto 3C3 horizon. However, afterwards decreasing trend with the depth was observed.

From surface Ap horizon upto 2Bt2 horizon cation exchange capacity values showed an increasing trend with depth in pedon IV. Afterwards regular decreasing trend with the depth was observed in this profile.

The pedon V, highest cation exchange capacity was observed in Bt2 (20.01 c.mol (p^+) kg^{-1}) horizon and lowest cation exchange capacity was observed in 2Bt5 horizon. From surface Ap horizon, there is an increase in cation exchange capacity till Bt2 horizon, afterwards regular decreasing trend was observed with depth.

In pedon VI, lowest cation exchange capacity was noticed in surface horizon. Afterwards there is a regular increase in cation exchange capacity upto 2Bt4 horizon and in the last horizon (2Bt5 horizon) it was 10.05 c.mol (p^+) kg^{-1}.

4.3.4.1 Calcium

Calcium was the most dominant exchangeable cation observed in all the pedons.

In pedon I, calcium content varied between 5.17 and 12.48 c.mol (p^+) kg^{-1} in Ap and Bt2 horizon, respectively. This indicated increasing trend with the depth, however afterwards it showed a decreasing trend with the depth.

Highest amount of calcium content was observed in Bt1 [8.42 c.mol (p^+) kg^{-1}] in pedon II. In surface and Bt1 horizon the calcium content was higher, afterwards it decreased with depth.

In pedon III the lowest calcium content [4.95 c.mol (p^+) kg^{-1}] was observed in the 3C5 horizon, whereas it was highest in the 3C2 horizon [14.32

c.mol (p^+) kg^{-1}]. From surface horizon, it was increasing with depth upto the 3C2 horizon and afterwards a decreasing trend in calcium content was noticed in the lower horizons.

In the IV pedon, from the beginning of surface horizon increasing trend with depth was observed upto 2Bt2 horizon where calcium content was found to be 14.00 c.mol (p^+) kg^{-1}. In the last three horizons, regular decreasing trend was observed.

In V pedon highest calcium content was present in Bt1 horizon i.e., 12.28 c.mol (p^+) kg^{-1}. Whereas, lowest content was observed in the lower most 2Bt5 horizon. However, there was regular decreasing trend with the depth in the profile.

In pedon VI, calcium content was found to vary between 5.10 c.mol (p^+) kg^{-1} (surface horizon) and 10.59 c.mol (p^+) kg^{-1} (2Bt3 horizon), which indicated increasing trend with depth.

4.3.4.2 Exchangeable magnesium

Magnesium was found to be the most dominant exchangeable cation, next to calcium in all the study profiles. Exchangeable magnesium content of pedon I varied between 1.81 and 3.49 c.mol (p^+) kg^{-1} in Ap and Bt2 horizons, respectively which indicated slight increase with the depth. Whereas in the last two horizons, it was found to be 2.14 and 2.53 c.mol (p^+) kg^{-1}, respectively.

In pedon II, magnesium content (1.41 c.mol (p^+) kg^{-1}) was highest in Bt1 horizon, afterwards decreasing trend with the depth was observed. Lowest magnesium content (1.45 c.mol (p^+) kg^{-1}) was present in last horizon i.e., 2C horizon.

In the III pedon, as the depth of the profile increases there was increase in the magnesium content i.e., 2.51 to 3.11 c.mol (p^+) kg^{-1} in Ap and BW2, respectively. However slight decrease in the magnesium content was observed in the last four successive horizons.

Magnesium content showed increasing trend with the depth of the profile from surface horizon to 2Bt2 horizon (3.08 to 4.36 c.mol (p^+) kg^{-1}) in pedon IV. In the last three horizons magnesium content was found to be decreased from 2.17 to 1.40 in BC and C horizons.

Highest magnesium content 4.63 c.mol (p^+) kg^{-1}) was observed in 2Bt2 horizon whereas lowest magnesium content was present in the surface horizon (0.78 c.mol (p^+) kg^{-1}) indicating a decreasing trend with depth in pedon V.

Exchangeable magnesium content was found to be varied between 1.56 c.mol (p^+) kg^{-1} and 4.99 c.mol (p^+) kg^{-1} in Bt2 and 2Bt5 horizons. Same amount of magnesium content (1.02 c.mol (p^+) kg^{-1}) was present in Bt3 and 2Bt4 horizons. However, an increasing trend with depth was observed in pedon VI.

4.3.4.3 Exchangeable potassium

Exchangeable potassium content was found to be low in all the profiles. Out of the six horizons, first three horizons showed increasing trend with the depth whereas the lower horizons showed a decreasing trend with the depth in pedon I. The potassium content was low (0.20 c.mol (p^+) kg^{-1}) in 2Bt4 horizon and high (0.53 c.mol (p^+) kg^{-1}) in Bt2 horizon in pedon I.

In pedon II, the surface horizon contained 0.35 c.mol (p^+) kg^{-1} of potassium and C horizon contained lowest (0.29 c.mol (p^+) kg^{-1}) potassium content. In the profile, regular decreasing trend in potassium content with depth was observed.

A regular increasing trend of potassium content was observed from surface horizon to 3C2 horizon (0.56 c.mol (p^+) kg^{-1} and 0.68 c.mol (p^+) kg^{-1}) in pedon III.

In pedon IV, surface horizon contained highest amount of exchangeable potassium (0.78 c.mol (p^+) kg^{-1}) and the lowest amount was present in C horizon (0.19 c.mol (p^+) kg^{-1}). This indicated a regular decreasing trend with depth.

Exchangeable potassium content of pedon V varied between 0.37 and 0.67 c.mol (p^+) kg^{-1} c.mol (p^+) kg^{-1} in C horizon and Bt3 horizon, indicated a regular decreasing trend with the depth in the profile.

In pedon VI, exchangeable potassium content varied between 0.22 c.mol (p^+) kg^{-1} and 0.53 c.mol (p^+) kg^{-1} in 2Bt5 horizon and 2Bt4 horizon. However, a regular increasing trend with the depth was observed in this profile.

4.3.4.4 Exchangeable sodium

Exchangeable sodium content was found to be low in all the profiles studied. In pedon I, increasing trend with the depth of the profile was observed upto Bt2 horizon, where exchangeable sodium content was highest (0.68 c.mol (p^+) kg^{-1}).

Exchangeable sodium content was highest in first two horizons i.e., Ap (0.57 c.mol (p^+) kg^{-1}) and Bt1 (0.54 c.mol (p^+) kg^{-1}) afterwards a regular decreasing trend with the depth of the profile was observed in pedon II.

In pedon III exchangeable sodium content was highest in the 3C3 horizon (1.67 c.mol (p^+) kg^{-1}) and lowest in the BW1 horizon (0.41 c.mol (p^+) kg^{-1}). This indicated a irregular increase in exchangeable sodium with the depth in this profile.

In pedon IV, exchangeable sodium content varied between 1.25 c.mol (p^+) kg^{-1} and 0.25 c.mol (p^+) kg^{-1} in surface Ap horizon and lower most C horizon, respectively indicating a decreasing trend with the depth in this profile.

As the depth increases, there was an increase in exchangeable sodium content from surface Ap horizon to Bt2 horizon (0.71 c.mol (p^+) kg^{-1} to

Table 4 (contd.....)

Profile No.	Horizon	Depth (cm)	CEC c mol (p+) kg^{-1} soil	CEC cmol (p+) kg^{-1} clay	Exchangeable cations cmol (p+) kg^{-1}				Total exchangeable bases cmol (p+) kg^{-1}	Base saturation (%)
					Ca	Mg	K	Na		
01	Ap	0-19	8.35	23	5.17	1.81	0.21	0.39	7.58	91
	Bt1	19-44	12.26	25	10.07	2.82	0.41	0.66	13.56	80
	Bt2	44-68	19.11	38	12.48	3.49	0.53	0.68	17.18	90
	Bt3	68-104	15.01	28	10.64	3.08	0.31	0.60	14.69	98
	2Bt4	104-131	09.58	25	6.78	2.14	0.20	0.35	9.47	99
	2Bt5	131-164	11.68	30	8.32	2.53	0.28	0.41	11.54	99
02	Ap	0-25	10.21	31	6.88	2.19	0.35	0.57	9.09	98
	Bt1	25-57	12.91	23	8.42	2.55	0.61	0.54	12.12	94
	Bt2	57-75	10.87	21	6.12	1.84	0.44	0.27	8.67	80
	Bt3	75-99	11.93	22	6.36	1.85	0.44	0.27	8.92	75
	2Bt4	99-133	9.28	19	6.46	2.01	0.41	0.20	9.03	98
	2Bt5	133-180	7.09	17	4.61	1.47	0.31	0.11	6.50	92
	C	180-190	7.46	19	3.19	1.45	0.29	0.15	5.08	88

Table 4 (contd……)

Profile No.	Horizon	Depth (cm)	CEC cmol (p+) kg^{-1} soil	CEC cmol (p+) kg^{-1} clay	Exchangeable cations cmol (p+) kg^{-1}				Total exchangeable bases cmol (p+) kg^{-1}	Base saturation (%)
					Ca	Mg	K	Na		
03	Ap	0-17	13.52	46	9.33	2.51	0.56	0.43	12.83	95
	BW1	17-30	9.41	39	6.24	1.83	0.25	0.41	8.73	93
	BW2	30-68	17.17	36	11.61	3.11	0.55	1.02	16.29	95
	2C1	68-76	15.83	31	10.94	2.86	0.47	0.75	15.02	95
	3C2	76-99	19.60	37	14.32	2.88	0.68	1.31	19.19	98
	3C3	99-121	19.30	47	13.6	2.86	0.53	1.67	18.66	97
	3C4	121-158	16.80	43	10.71	3.25	0.59	1.57	16.12	96
	3C5	158-178	8.08	25	4.95	8.03	0.30	0.61	7.89	98
04	Ap	0-18	16.63	38	11.00	3.08	0.78	1.25	16.11	97
	Bt1	18-38	17.77	41	11.66	3.55	0.69	1.15	17.05	96
	2Bt2	38-66	20.37	38	14.00	4.36	0.72	1.00	20.08	99
	2Bt3	66-92	14.44	37	9.73	2.97	0.48	0.52	13.70	95
	2Bt4	92-126	9.35	28	6.69	1.72	0.18	0.27	8.86	95
	Bc	126-155	8.67	27	5.70	2.17	0.31	0.46	8.64	100
	C	155-178	6.37	21	4.41	1.40	0.19	0.25	6.25	98

Table 4 (contd.....)

Profile No.	Horizon	Depth (cm)	CEC cmol (p+) kg^{-1} soil	CEC cmol (p+) kg^{-1} clay	Exchangeable cations cmol (p+) kg^{-1}				Total exchangeable bases cmol (p+) kg^{-1}	Base saturation (%)
					Ca	Mg	K	Na		
05	Ap	0-25	12.26	32	7.55	3.18	0.55	0.71	11.99	98
	Bt1	25-55	19.58	44	12.28	4.56	0.61	1.34	18.79	96
	Bt2	55-82	20.01	40	12.15	4.63	0.55	1.55	18.88	94
	Bt3	82-112	17.63	36	11.27	3.78	0.67	1.19	16.91	96
	2Bt4	112-149	16.30	42	8.53	3.84	0.45	1.18	14.00	86
	2Bt5	149-180	15.32	40	8.04	3.23	0.37	0.91	12.55	82
06	Ap	0-14	8.25	28	5.10	2.07	0.35	0.53	8.05	98
	Bt1	14-41	11.42	30	7.15	3.02	0.34	0.79	11.30	99
	Bt2	41-75	18.96	35	10.39	4.99	0.46	1.58	17.47	92
	2Bt3	75-111	17.66	34	10.59	2.93	0.51	1.49	15.52	88
	2Bt4	111-142	19.42	35	10.43	3.67	0.53	1.47	16.10	83
	2Bt5	142-162	10.05	22	5.76	1.56	0.22	0.48	8.02	80

1.55 c.mol (p^+) kg^{-1}) afterwards regular decreasing trend with the depth was observed in pedon V.

In pedon VI, exchangeable sodium content was found to be varied between 0.48 c.mol (p^+) kg^{-1} and 1.58 c.mol (p^+) kg^{-1} in 2Bt5 and Bt2 horizons.

4.3.5 Available nutrients

The data related to available nitrogen, phosphorus, potassium and micronutrients are presented in Table 6.

4.3.5.1 Available nitrogen

The available nitrogen content of the soils was found to vary from low to medium in all the pedons. In pedon I it varied between 170.75 kg ha^{-1} in surface horizon and 28.98 kg ha^{-1} in 2Bt3 horizon, respectively, indicating low nitrogen content and decreasing trend with the depth. Subsequently increased in 2Bt4 and 2Bt5 horizons (131.89 and 143.27 kg ha^{-1}, respectively

In pedon II, highest nitrogen content (172.51 kg ha^{-1}) was present in Bt3 horizon. Whereas C horizon contained 96.18 kg ha^{-1}. Generally a decreasing trend with depth was observed in this profile.

A regular decreasing trend with the depth was observed in pedon II from 155.38 kg ha^{-1} in surface horizon to 72.40 kg ha^{-1} in 3C4 horizon.

In pedon IV, root zone (rhizosphere) contained highest available nitrogen (available) content i.e., 234.77 kg ha^{-1} (Bt1 horizon) whereas lowest amount of available nitrogen content was present in C horizon 56.51 kg ha^{-1}, indicating a decreasing trend throughout the depth of the profile.

In pedon V, available nitrogen content was found to be varied between 118.70 kg ha^{-1} in 2Bt5 horizon (lower most horizon) and 203.84 kg ha^{-1} in surface horizon, respectively, which indicated a decreasing trend with the depth in this profile.

Highest amount (207.40 kg ha^{-1}) of available nitrogen content was present in surface horizon in pedon VI, whereas lowest amount was found in 2Bt5 horizon (136.77 kg ha^{-1}) and exhibited a decreasing trend with the depth.

4.3.5.2 Available phosphorus (Bray's – P)

The available phosphorus content of the soils, estimated in Bray's I extract was found to fluctuate between low to medium in all the pedons studied which indicated a need to supplement the required deficit amount of phosphorus through phosphatic fertilizers.

Available phosphorus content of pedon I, varied between 24.88 kg ha^{-1} in 2Bt5 horizon and 19.42 kg ha^{-1} in surface Ap horizon indicating regular increasing trend with the depth.

Higher amount i.e., 22.84 kg ha^{-1} of available phosphorus content was present in pedon II, a significant difference in phosphorus (available) content in each horizon was noticed. Lower amount of available phosphorus i.e., 21.70 kg ha^{-1} was present in C horizon.

Available phosphorous content was found to be 23.29 kg ha^{-1} in 3C3 horizon and 10.38 kg ha^{-1} in 3C5 horizon, respectively in pedon III. However fluctuation in the available phosphorus content throughout the depth of the profile was observed.

In pedon IV, Bt1 horizon contained 28.22 kg available phosphorus per ha. Whereas, it was 16.15 kg per ha in C horizon. This indicates a decreasing trend with the depth.

The horizons Bt1, Bt2 and Bt3 contained 26.53, 25.87 and 23.29 kg ha^{-1} of available phosphorous content, respectively in pedon V. Whereas surface horizon contained 19.23 kg ha^{-1} and lower most 2Bt5 horizon contained 16.30 kg ha^{-1} of available phosphorous. Generally a decreasing trend with the depth was observed in this profile in available phosphorus.

In pedon VI, surface horizon contained highest amount of available phosphorus (34.27 kg ha^{-1}) and Bt3 horizon contained 10.38 kg ha^{-1} of phosphorous (available) which indicated a regular decrease in phosphorous (available) content with the depth.

4.3.5.3 Available potassium

The available potassium content of the soils estimated in NH_4OAc extract was noted to fluctuate between low to medium in all the pedons thereby indicating the need to supplement the necessary deficit quantity.

The available potassium content of pedon I was found to be highest in Bt1 horizon i.e., 128.68 kg ha^{-1} whereas lowest was present in 2Bt4 horizon i.e., 87.36 kg ha^{-1}. From surface horizon upto Bt2 horizon increasing trend was observed, afterwards decreasing trend was observed in this profile.

In pedon II, lowest amount of available potassium was present in C horizon (6.46 kg ha^{-1}) where as highest amount was present in surface horizon (215.15 kg ha^{-1}), which indicated a regular decreasing trend with the depth of the profile.

177.18 kg ha^{-1} of available potassium was present in surface horizon in pedon III and 111.36 kg ha^{-1} of potassium (available) was present in 2C1 horizon, indicating a decreasing trend with the depth in this profile.

In pedon IV, surface horizon contained highest amount (227.47 kg ha^{-1}) of available potassium content whereas lowest amount was present in C horizon (52.08 kg ha^{-1}) indicating a regular decreasing trend with the depth in this profile.

As the depth of the profile increases there was decrease in available potassium content from 165.76 kg ha^{-1} in surface horizon and 95.20 kg ha^{-1} in 2Bt5 horizon, clearly indicating a regular decreasing trend with the depth in profile V.

Pedon VI, exhibited a decreasing trend like other profiles where 258.60 kg ha^{-1} of available potassium was present in surface horizon and lower most horizon contained 132.16 kg ha^{-1} of available potassium.

4.3.5.4 Available micronutrients

The data pertaining to the available micronutrients are presented in Table 6. The available micronutrients viz., DTPA extractable iron, copper manganese, zinc at pH 7.3 of the soils were observed to fluctuate between deficient to optimum level in all the pedons.

4.3.5.4.1 DTPA extractable Fe

DTPA extractable iron content was found to be varied among the horizons. It was 0.94 ppm in surface horizon, which increased to 1.33 ppm in Bt2 horizon and afterwards decreased regularly till the last 2Bt5 horizon (0.45 ppm) in pedon I.

A regular decreasing trend was observed in pedon II. Surface horizon contained 2.18 ppm whereas in lower most C horizon 0.48 ppm of DTPA extractable iron was present.

In pedon III, highest amount of DTPA extractable iron was present in BW1 horizon i.e., 2.30 ppm and showed decreasing trend regularly with the depth except surface horizon had the lowest content 0.44 ppm of iron (DTPA extractable).

In pedon IV, 2.82 ppm of DTPA extractable iron was present in 2Bt2 horizon, afterwards it showed a decreasing trend with the depth. 2.33 ppm and 0.93 ppm of iron (DTPA extractable) was present in Ap (surface) horizon and C horizon, respectively.

Between 2.82 ppm and 0.96 ppm of DTPA extractable iron was present in surface horizon and lower most 2Bt5 horizon, confirming a regular decreasing trend in iron content with the depth of profile V.

In pedon VI, higher concentration i.e., 4.23 ppm of DTPA extractable iron was present in surface horizon. However, it showed a regular decreasing trend with depth.

4.3.5.4.2 DTPA extractable Cu

DTPA extractable copper content in pedon I, varied between 0.38 ppm in surface horizon and 0.07 ppm in lower most 2Bt5 horizon indicating a regular decreasing trend with the depth.

In pedon II, 0.49 ppm and 0.18 ppm were the DTPA extractable copper concentrations in surface Ap and lower most C horizons, respectively which clearly indicated a decreasing trend with the depth of the profile.

DTPA extractable copper content varied between 0.22 ppm in Ap horizon and 0.09 ppm in 3C4 horizon, respectively in pedon III, indicating a decreasing trend with the depth.

A regular decreasing trend with depth was observed in pedon IV, where copper (DTPA extractable) content was reduced from 0.58 ppm in surface horizon to 0.08 ppm in lower most C horizon, respectively.

In pedon V, DTPA extractable copper content varied between 0.39 ppm (surface horizon) and 0.09 ppm (lower most horizon) which clearly indicated a regular decrease in copper content. Three successive horizons Bt3, 2Bt4 and 2Bt5 contained equal concentrations of DTPA extractable copper i.e., 0.12 ppm in pedon VI, whereas in surface horizon 0.41 ppm of copper (DTPA extractable) content was present.

4.3.5.4.3 DTPA extractable Mn

The DTPA extractable manganese content of the soils in pedon I was found to be between 6.84 ppm and 1.03 ppm in surface horizon and C horizon, respectively confirming presence of adequate amount of manganese in this profile, which indicates decreasing trend with the depth.

In pedon II, surface horizon contained higher amount i.e., 9.01 ppm of DTPA extractable manganese, whereas C horizon contained 1.02 ppm clearly indicating a regular decrease in manganese (DTPA extractable) content with depth in the profile.

In pedon III 3C5 horizon contained 0.22 ppm of DTPA extractable manganese content and Ap horizon contained 5.96 ppm of manganese (DTPA extractable) indicating a decreasing trend with the depth.

In pedon IV, adequate quantity of DTPA extractable manganese was present in surface horizon i.e., 8.12 ppm where as lower most horizon C, contained only 2.75 ppm, which conferred the decreasing trend with depth of the profile.

In pedon V, DTPA extractable manganese content ranged between 8.20 ppm in Ap horizon and 2.75 ppm in 2Bt5 horizon indicating adequate manganese status. However, a regular decrease from surface to subsurface horizons and high manganese content in surface horizon was noticed.

In pedon VI, DTPA extractable manganese content varied between 8.96 ppm in surface Ap horizon and 0.14 ppm in lower most 2Bt5 horizon, respectively which indicated a decreasing trend with the depth.

4.3.5.4.4 DTPA extractable Zn

The DTPA extractable zinc content of the profile I varied between 1.01 ppm in Bt1 horizon and 0.36 ppm in 2Bt5 horizon, respectively. However adequate amount of zinc (DTPA extractable) content was present in surface horizon.

The profile II, 0.86 ppm of zinc content was observed in surface horizon and it decreased regularly in subsurface horizons to 0.37 ppm in C horizon.

In pedon III, zinc content varied between 0.49 ppm and 0.14 ppm in Ap and 3C5 horizons, respectively. However, this indicated a regular decreasing trend with depth of profile.

Surface horizon of pedon IV contained 0.79 ppm, which decreased progressively to 0.13 ppm in 2Bt4 which remained static in the subsequent horizons BC and C.

In pedon V, DTPA extractable zinc content ranged between 0.46 ppm in Ap horizon and 0.13 ppm in 2Bt5 horizon, which indicated decreasing trend with the depth of the profile.

In pedon VI, subsurface Bt1 horizon contained higher amount of 1.04 ppm of Zn i.e., 0.20 ppm was present in lower most 2Bt5 horizon. This indicated regular decreasing trend with the depth.

Table 5. Available nutrients content of the pedons

Profile No.	Horizon	Depth (cm)	Av.N (kg ha^{-1})	Av.P (kg ha^{-1})	Av.K (kg ha^{-1})	Av. Micronutrients (ppm)			
						Fe	Mn	Cu	Zn
01	Ap	0-19	170.75	19.42	107.58	0.96	6.84	0.38	0.97
	Bt1	19-44	105.21	17.53	128.68	1.26	3.89	0.33	1.01
	Bt2	44-68	81.98	16.54	126.33	1.33	3.92	0.29	0.99
	Bt3	68-104	81.98	25.94	110.54	1.03	2.19	0.19	0.83
	2Bt4	104-131	131.89	26.01	87.36	0.85	1.58	0.08	0.73
	2Bt5	131-164	143.27	24.88	91.95	0.45	1.03	0.07	0.36
02	Ap	0-25	162.91	22.84	215.15	2.18	9.01	0.49	0.86
	Bt1	25-57	171.56	19.42	180.04	1.60	6.31	0.28	0.80
	Bt2	57-75	168.68	15.99	132.83	1.30	4.38	0.30	0.60
	Bt3	75-99	172.57	23.33	121.63	0.92	3.31	0.27	0.46
	2Bt4	99-133	168.57	22.84	85.12	0.86	2.48	0.27	0.35
	2Bt5	133-180	106.17	21.91	74.92	0.74	1.82	0.19	0.53
	C	180-190	96.18	21.70	6.46	0.48	1.02	0.18	0.37

Table 5 (contd....)

Profile No.	Horizon	Depth (cm)	Av.N (kg ha^{-1})	Av.P (kg ha^{-1})	Av.K (kg ha^{-1})	Av. Micronutrients (ppm)			
						Fe	Mn	Cu	Zn
03	Ap	0-17	155.38	19.42	177.18	0.44	5.96	0.22	0.49
	BW1	17-30	137.24	21.70	115.02	2.30	2.56	0.18	0.37
	BW2	30-68	78.75	13.97	139.66	1.80	1.48	0.19	0.31
	2C1	68-76	85.62	15.83	111.36	0.90	1.03	0.16	0.26
	3C2	76-99	108.05	22.34	173.71	0.68	1.00	0.16	0.17
	3C3	99-121	96.45	23.29	154.11	0.59	0.68	0.11	0.15
	3C4	121-158	72.40	15.14	153.88	0.45	0.54	0.09	0.15
	3C5	158-178	78.08	10.38	156.88	0.26	0.22	0.30	0.14
04	Ap	0-18	209.16	21.91	227.47	2.33	8.12	0.58	0.79
	Bt1	18-38	234.77	28.22	129.13	2.70	5.78	0.48	0.64
	2Bt2	38-66	223.89	27.04	109.08	2.82	4.94	0.32	0.58
	2Bt3	66-92	168.89	23.52	109.36	2.04	4.06	0.26	0.49
	2Bt4	92-12	111.09	27.30	75.93	0.03	4.12	0.23	0.13
	BC	126-155	94.33	18.63	64.06	0.95	3.78	0.09	0.13
	C	155-178	56.51	16.15	52.08	0.93	2.75	0.08	0.13

Table 5 (contd.....)

Profile No.	Horizon	Depth (cm)	Av.N (kg ha^{-1})	Av.P (kg ha^{-1})	Av.K (kg ha^{-1})	Av. Micronutrients (ppm)			
						Fe	Mn	Cu	Zn
05	Ap	0-25	203.84	19.23	165.76	2.82	8.20	0.39	0.46
	Bt1	25-55	194.78	26.53	129.58	1.83	4.50	0.40	0.41
	Bt2	55-82	189.84	25.87	107.74	1.60	3.72	0.24	0.32
	Bt3	82-112	177.60	23.29	101.36	1.30	3.48	0.20	0.28
	2Bt4	112-149	176.90	17.47	73.58	1.14	2.96	0.12	0.28
	2Bt5	149-180	118.70	16.30	95.20	0.96	2.75	0.09	0.13
06	Ap	0-14	207.40	34.27	258.60	4.23	8.96	0.41	0.88
	Bt1	14-41	175.78	19.61	99.90	2.80	4.95	0.28	1.04
	Bt2	41-75	188.83	17.30	83.66	1.96	2.56	0.20	0.83
	2Bt3	75-111	181.83	13.38	116.25	1.73	1.99	0.12	0.64
	2Bt4	111-142	172.25	11.53	80.52	1.68	0.89	0.12	0.36
	2Bt5	142-162	136.77	17.47	132.16	1.30	0.14	0.12	0.20

DISCUSSION

V. DISCUSSION

Results of the analysis of the six selected pedons presented in Chapter IV are discussed under the following aspects:

5.1 Morphological characteristics of soils.

5.2 Physical characteristics of soils.

5.3 Chemical characteristics of soils.

5.4 Classification of soils.

5.5 Available nutrient status of soils.

5.1 Morphological characteristics of soils

The pedon I located in the upland region (3-5 per cent slope) of Neralaghatta village extends to a depth of 164 cm, indicating very deep solum depth. The colour of the soil varied from dark reddish brown (moist) to light reddish brown moist. Soil texture varied between clay in the first four horizons to clay loam in the last two horizons, with medium weak to moderate sub-angular blocky structure throughout the depth of the profile. The coarse fragments (per cent by volume) varied from 5-10 per cent upto the depth of 44 cm and afterwards it increased gradually to 70 per cent or more down the profile. Consistency of soils ranged from firm, slightly sticky in surface horizons to friable slightly plastic throughout the depth of the profile, porosity of soils varied from fine medium and few common in surface horizon to fine and many in subsurface horizons indicating more number of micropores (at

subsurface horizons). The soil was found to exhibit near neutral to slightly acidic soil reaction.

The pedon II was located in the midland region (1 to 3 per cent slope) of Neralaghatta village and it was found to be very deep (190 cm) and showed presence of clay coatings in Bt1, Bt2 and Bt3 horizons, indicating accumulation of clay in the subsurface horizons. Colour of the soil ranged from dark reddish brown to dark red and found to be mainly influenced by the parent material and low organic carbon content. Change in texture from sandy clay loam in surface horizon to clay in next three successive subsurface (Bt1, Bt2 and Bt3) horizons and clay loam in the last two horizons indicated accumulation of clay in the sub soil. The structure was found to be sub-angular blocky throughout the depth of the profile. Less than 5 per cent by volume of coarse fragments were present in subsurface horizon and increased gradually to 70 per cent and more. Friable, slightly plastic nature of consistency was observed throughout the profile (from surface to subsurface horizons). Porosity of these soils varied from fine medium to many throughout the depth of profile was noticed which in turn indicated prevalence of plenty of micropores in this pedon.

The third pedon is located in the nearly lowland region (0 to 1% slope) of Neralaghatta village. The depth of the solum was 178 cm with weak expression of profile development in the sub soil upto 68 cm depth. The effect of sedimentation is seen after 68 cm depth. The almost abrupt decrease in the

coarse sand content from fourth horizon onwards which continued upto 99 cm depth and its sudden increase later indicate the sedimentary nature of the material. Further the erratic distribution of fine sand, silt and clay in the subsoil clearly shows the effect of sedimentation in pedon III. The colour of the soil, which ranged from dark brown (7.5 YR 3/2) to very dark grayish brown (10 YR 3/2) also indicate the depository nature of the parent material. Negligible amount of coarse fragments was present throughout the depth. Consistency of soils was friable, slightly sticky in first four successive horizons upto a depth of 76 cm followed by very fine, very sticky, and very plastic in all the four successive horizons below. Porosity of soils varied from fine and common in the surface horizon to very fine and common throughout the depth of the profile in the subsoil. Structure of soil was found to be sub-angular blocky in first four horizons and massive wedge shaped aggregates in last four horizons. The reaction of the soil ranged from near neutral to acidic.

In pedon IV, the soils were found to be very deep with solum depth of 178 cm. This pedon is located in the upland region of Neralaghatta village (3 to 8% slope). Illuviation of clay was observed indicating the presence of argillic horizon. The colour of the soil was dark reddish brown and found to be influenced mainly by parent material. Texture of the soil changed from clay in the first three horizons to sandy clay loam in the subsequent horizons. Whereas, structure remained same throughout the depth of profile as weak to moderate, medium sub-angular blocky. In the surface horizon coarse

fragments were less than 10 per cent, which later increased abruptly to 70 per cent and remained more or less constant throughout the depth of the profile. Consistency of the soils of this pedon varied between friable and slightly plastic in surface horizon to very friable, very sticky and plastic in subsurface horizons. Fine, medium to many pores were present throughout the depth of the profile. Soils were found to be slightly acidic in soil reaction.

The pedon V located in the gently sloping region (3 to 5% slope) of Kestur village. This pedon was found to be very deep with gradual increase in clay content from surface horizon to subsurface horizons indicating translocation of clay in the soil. Colour of the soil changed from dark reddish brown to reddish brown showing the influence of parent material, less organic material prevalence and high temperature (22°C to 30°C) . Texture of the soil changed from sandy clay (in surface horizon) to clay (in subsurface horizons) indicating illuviation of clay. Structure of the soils remained constant throughout the depth of the pedon. No coarse fragments were observed in the first four horizons, whereas in the last two successive horizons fine gravel of 40 to 50 per cent was present. Friable and slightly plastic type of consistency remained unaltered throughout the depth of the profile. Fine and many pores were present in the major portion of the profile except in the surface horizon where fine, medium and many pores were noticed. Soils exhibited slightly acidic to near neutral soil reaction.

The sixth profile was located in the upland region (3 to 8% slope) of Kestur village. The solum depth was 162 cm indicating very deep nature of soil. The texture of the soil changed from sandy clay loam (in surface horizon to clay (in subsurface horizons) which indicates there was movement of clay particles by the percolating water from surface to subsurface horizons (Illuvial zone). Colour of the soil ranged between dark reddish brown to reddish brown which clearly showed the profound influence of free iron oxide and dominance of iron. Structure was found to be sub-angular blocky and remained unaltered throughout the depth of the profile. Per cent of fine coarse gravel of size 2.5 to 7.5 cm varied between 30 to 50 per cent throughout the depth of the profile except at surface horizon where only 10 per cent fine gravel was present. Consistency of these soils varied between friable, slightly sticky, and slightly plastic throughout the depth of the profile. Fine common to very fine common, medium few pores were observed in this profile in all the horizons. Soils were found to be acidic to near neutral in soil reaction. The chemical weathering was found to be quite highly pronounced in this profile.

Similar observations were made by Sahu *et al.* (1990), while studying morphology, genesis, mineralogy and classification of soils of northern plateau zone of Orissa.

5.2 Physical characteristics of soils

5.2.1 Bulk density

Bulk density of profile samples (samples from 0-25 cm) was determined by core sampler method. The bulk density of profile varied from 1.57 g cc^{-1} in pedon VI to 1.49 g cc^{-1} in pedon IV, which were collected within 25 cm from surface horizon. The values mentioned in Table 3 indicate the porous nature of soils. The ideal bulk density value for agricultural soils is 1.65 g cc^{-1} and since bulk density values are almost nearer to this value, these soils facilitate better root growth.

5.2.2 Particle size distribution

The data on the particle size distribution (Table 3 and Fig.10.1 to 10.6) indicated that the textural class of the soils in pedon I was clay to clay loam which indicated the movement of clay as clay per cent increased down the profile and there was reduction in sand content with depth. In pedon II, the particle size distribution indicated dominance of clay content in subsurface horizons with low silt content and appreciable sands content. Clay content increased down the profile till Bt3 horizon, with a simultaneous decrease in sand content there by suggesting illuviation of clay. This observation is in accordance with the results of Sahu *et al.* (1990) on red soils of Orissa. The examination of data on particle size distribution indicated that the textural class of these soils varied from sandy clay loam to clay. Reason for the occurrence of the above soil properties may possibly be attributed to the fact that these soils are formed on coarse grained granite gneiss parent material. Same

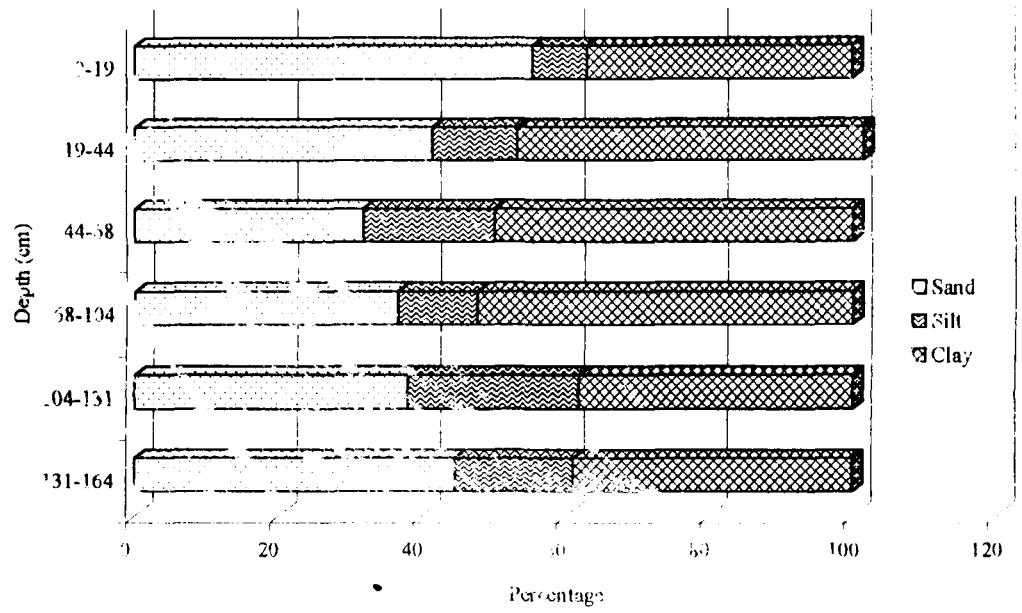


Fig.10.1: Particle size distribution of Pedon 1.

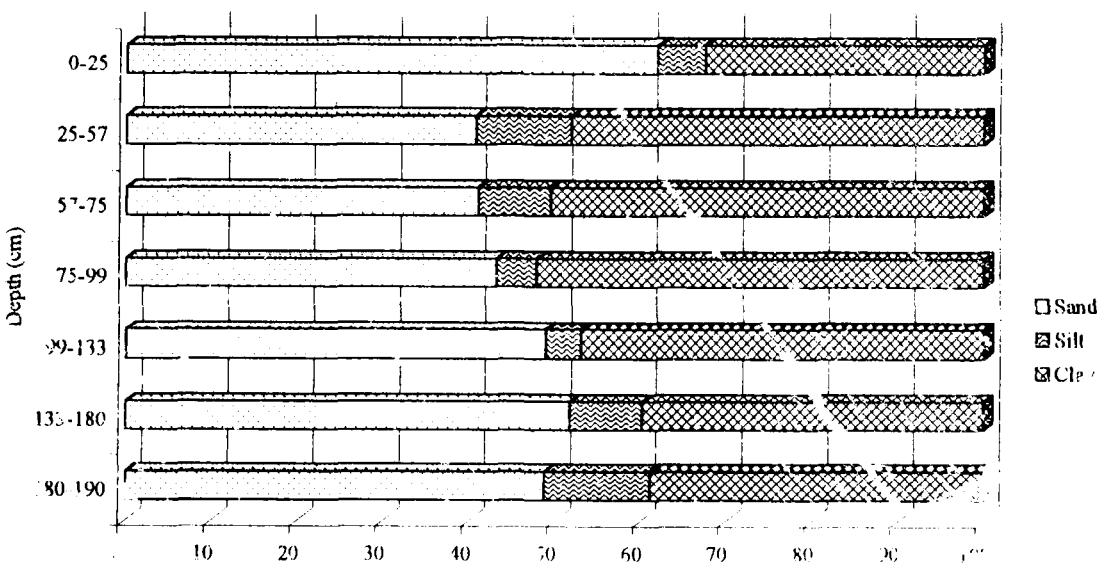


Fig.10.2: Particle size distribution of Pedon 2

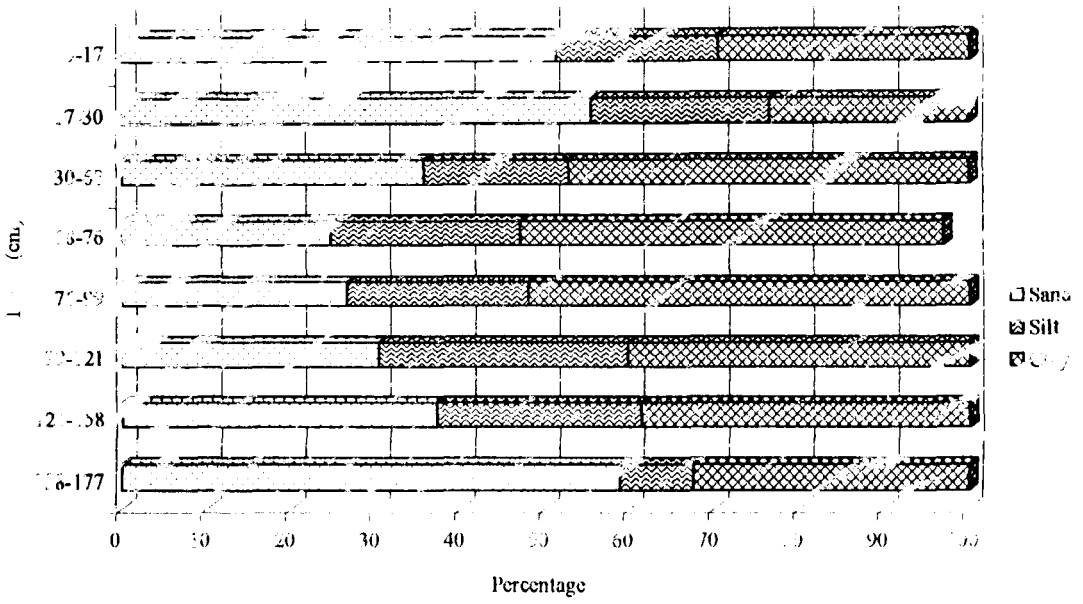
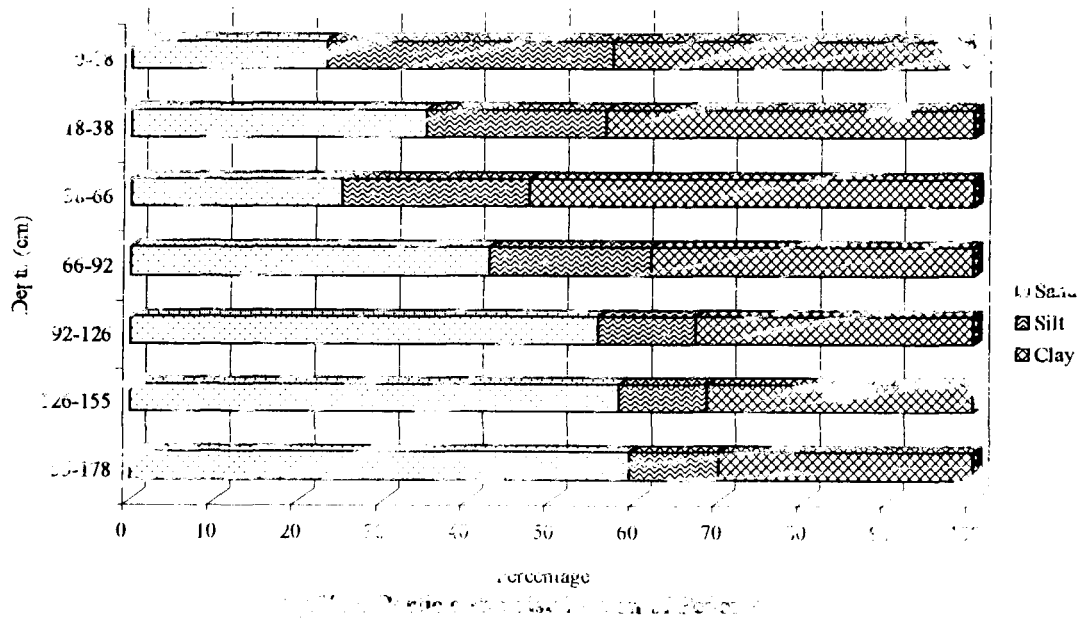


Fig. 10.3: Particle size distribution of Pedon 3



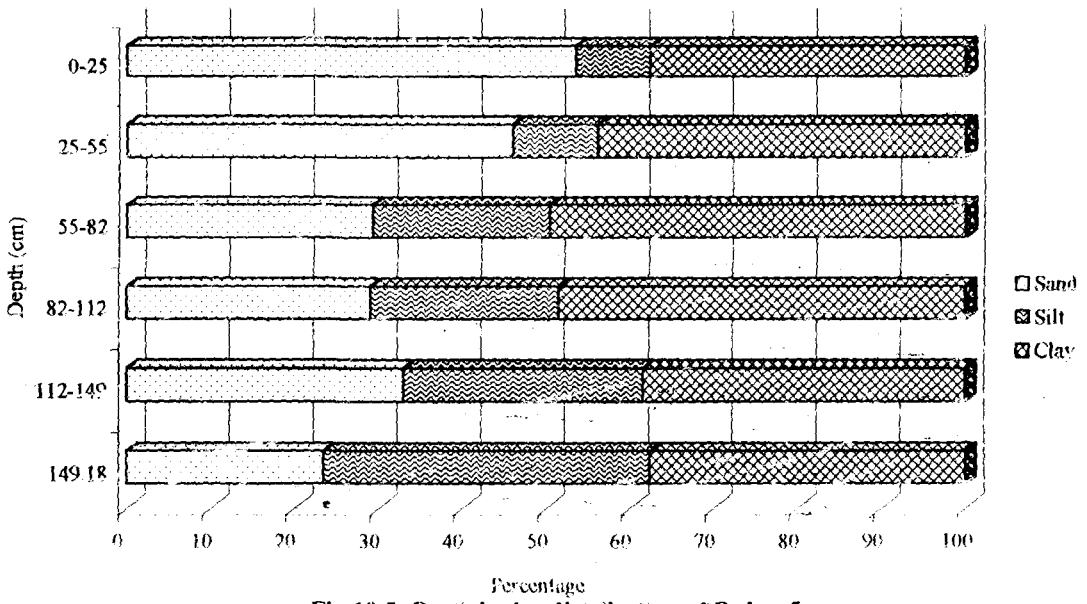


Fig.10.5: Particle size distribution of Pedon 5

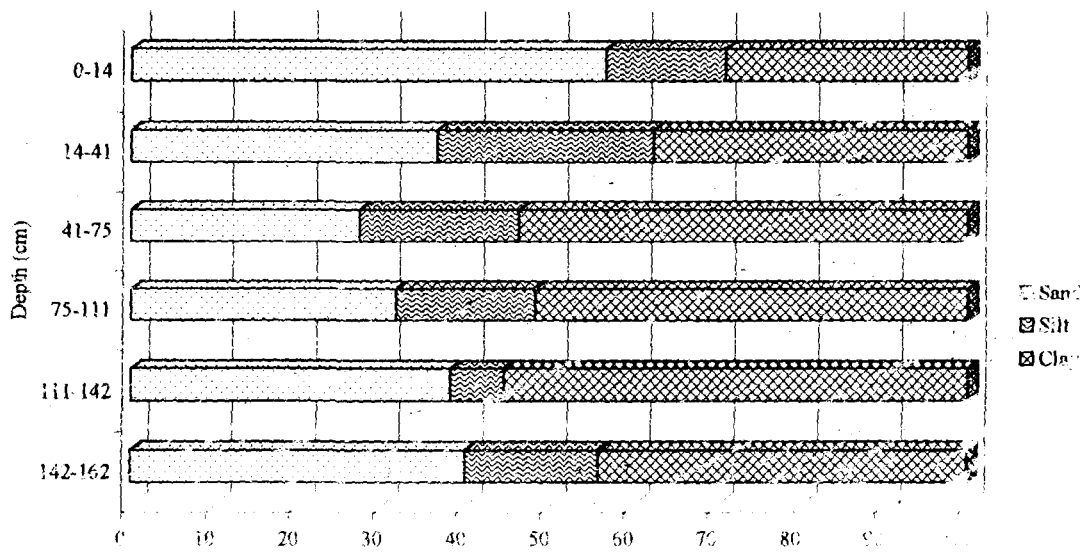


Fig.10.6: Particle size distribution of Pedon

discussion holds good for other remaining pedons also. These observations are further supported by the observations made by Bhaskar and Subbaiah (1995).

5.3 Chemical characteristics of soils

5.3.1 Soil reaction (pH)

The examination of data presented in the Table 4 and Fig.11.1 to 11.3 indicated that the soils of all the pedons were slightly acidic to near neutral in nature and pH showed an increasing trend with the depth of the soil. Since the pedons collected were from an area receiving low rainfall and did not have enough vegetative cover, acidic pH of the soil might be attributed to the acidic parent material which is a granite gneiss and known to be deficient in basic materials. This is in accordance with the results of Ananthanarayana and Perur (1973). The dominance of Kaolinite clay (1 : 1 type), low CEC and low bases have also significantly contributed for the occurrence of the acidic nature of soils in the study area.

5.3.2 Electrical conductivity

The examination of electrical conductivity values of soil (Table 4) of all the pedons estimated in 1 : 2.5 soil : water extract at 25°C revealed that the electrical conductivity was very low in all the soils, indicating prevalence of non-saline condition. This is due to the undulating nature of the terrain coupled with free drainage conditions, which favours removal of released bases by the percolating and drainage waters. This is in accordance with the similar observations made by Sahu and Dash (1993).

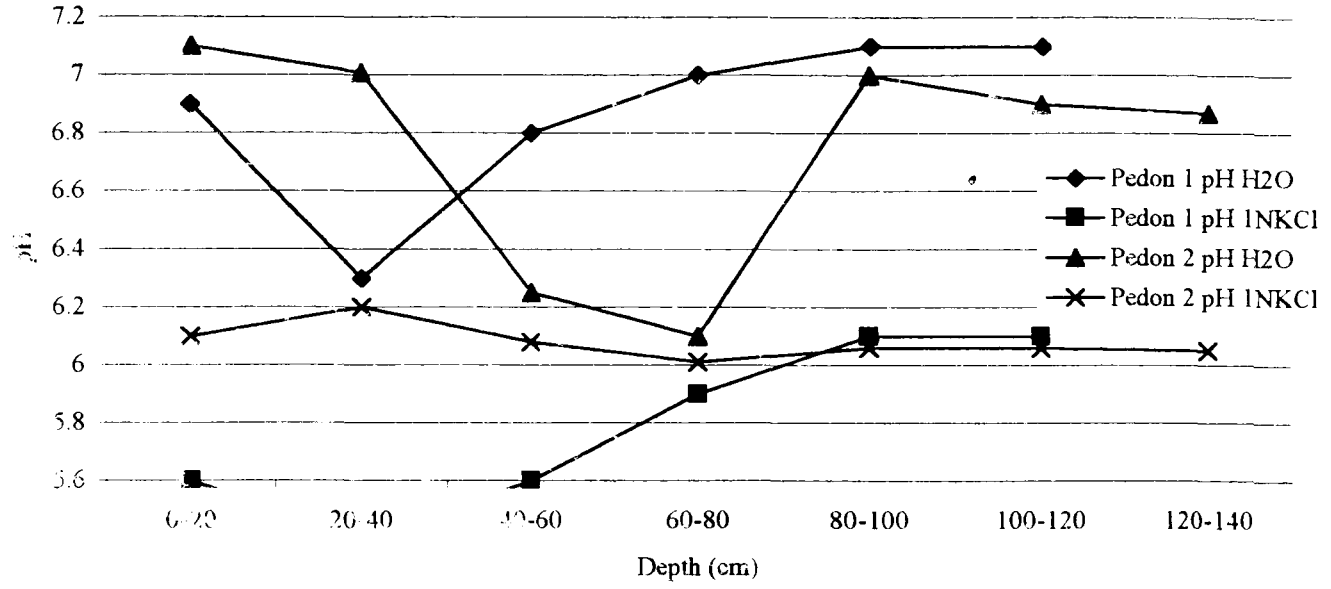


Fig. 11.1: Soil reaction classes of Pedons 1 and 2

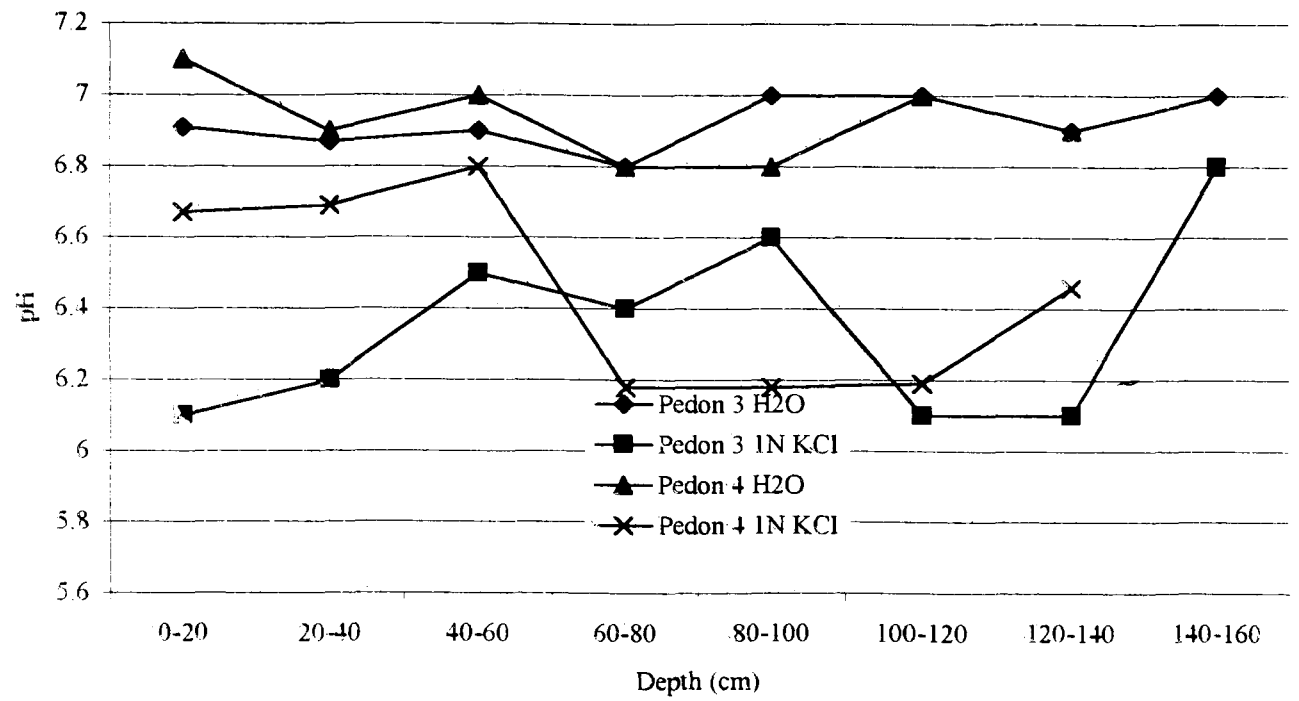


Fig.11.2: Soil reaction classes of Pedons 3 and 4

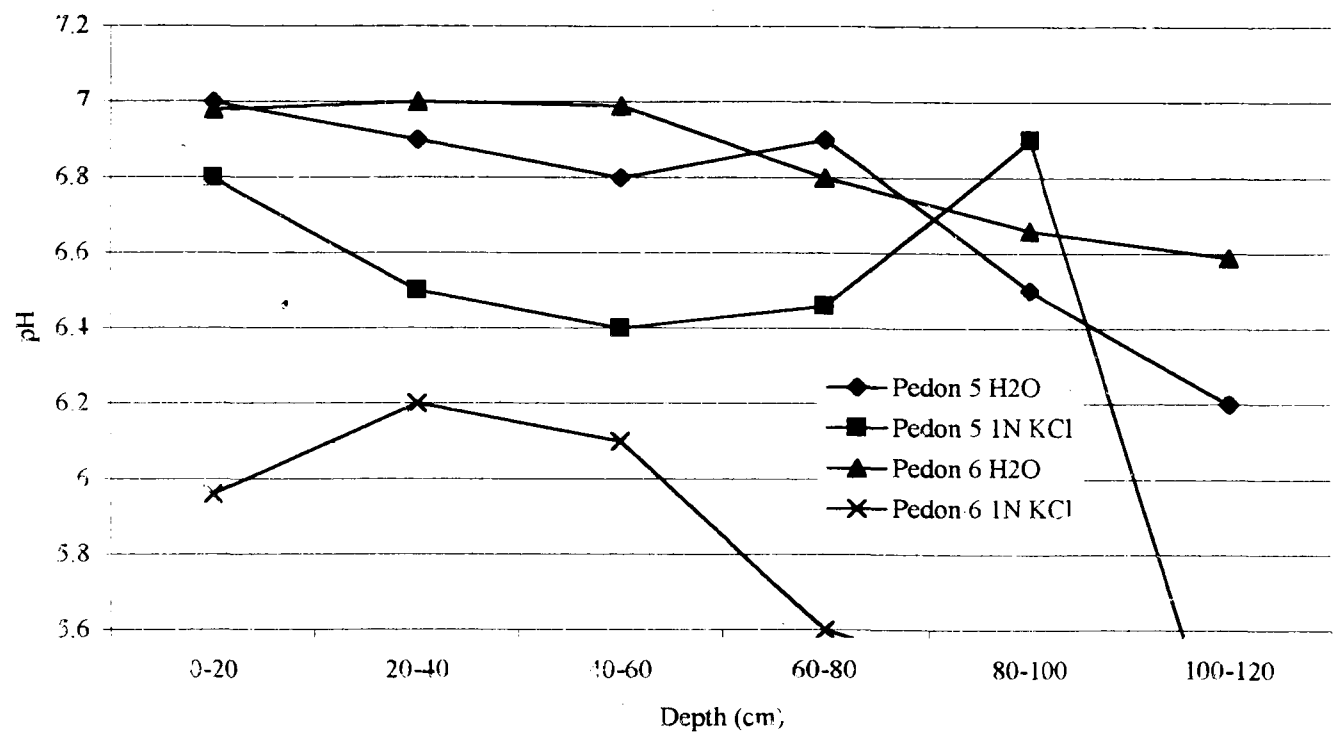


Fig.11.3: Soil reaction classes of Pedons 5 and 6

5.3.3 Organic carbon

The organic carbon content of soils (Table 4 and Fig.12) was found to vary between low to high. However, in all the pedons in general, organic carbon content of soils was found to fluctuate between low to medium levels and decreased regularly down the profile. These observations were in accordance with Krishnamoorthy and Govindarajan (1977). The reason for the low organic carbon content of the soils may be attributed to the prevalence of tropical conditions in the area. The organic matter degradation and removal takes place at a faster rate coupled with low vegetative cover, there by leaving less chances of accumulation of organic carbon in the soil. Besides the reasons sited above erosion also contributed for low organic carbon content accumulation in the soils of the study area.

5.3.4 Cation exchange capacity of soil

Data presented in Table 5 and Fig. 13.1 shows that cation exchange capacity of the soils in all the pedons estimated in NH_4OAc extract was found to vary between low to medium. Maximum cation exchange capacity was observed in the horizons of the pedons where illuviation or translocation of clay from surface to subsurface horizon has taken place, indicating the role of clay in the increased cation exchange capacity observed in the argillic horizons. However cation exchange capacity was found to be increasing regularly with depth in accordance with the increase in clay content and then decreasing regularly indicating a direct relationship with the clay per cent of the profile. These observations are in accordance with the results

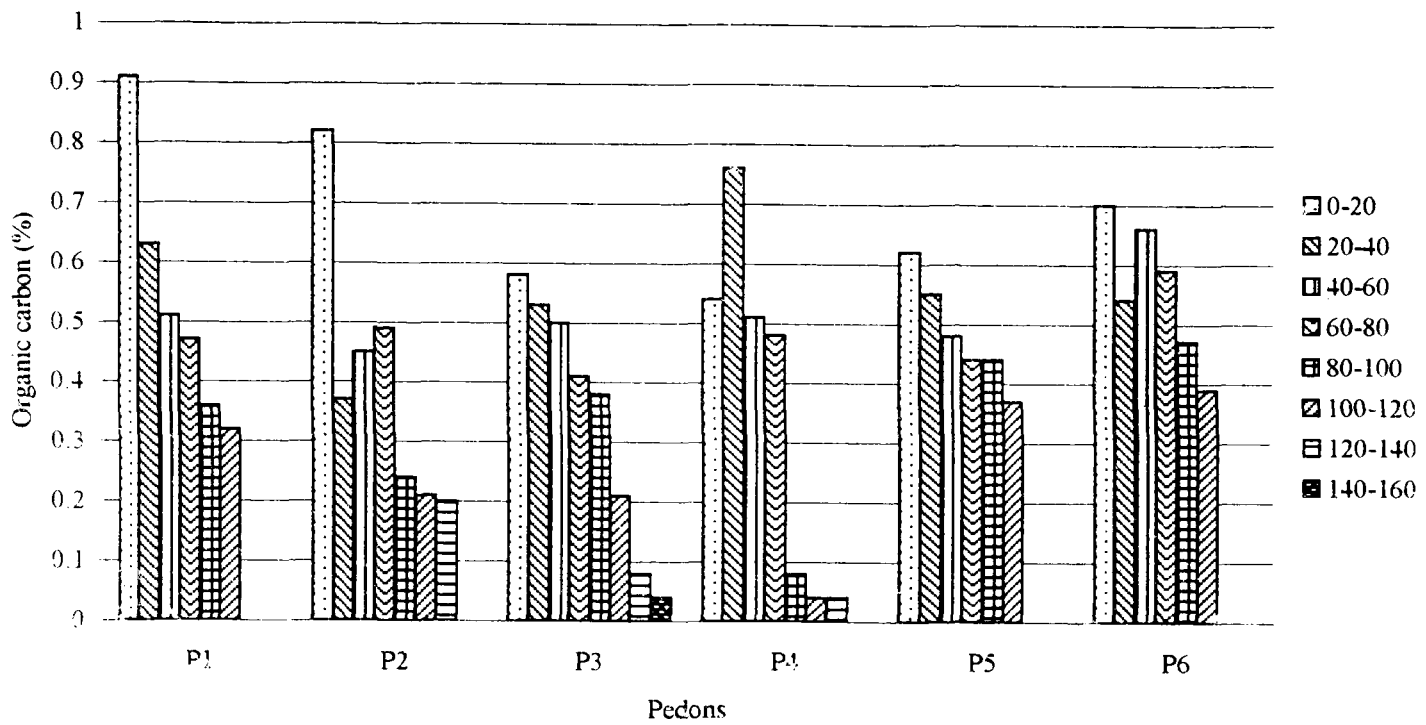


Fig 12: Organic carbon status of Pedons

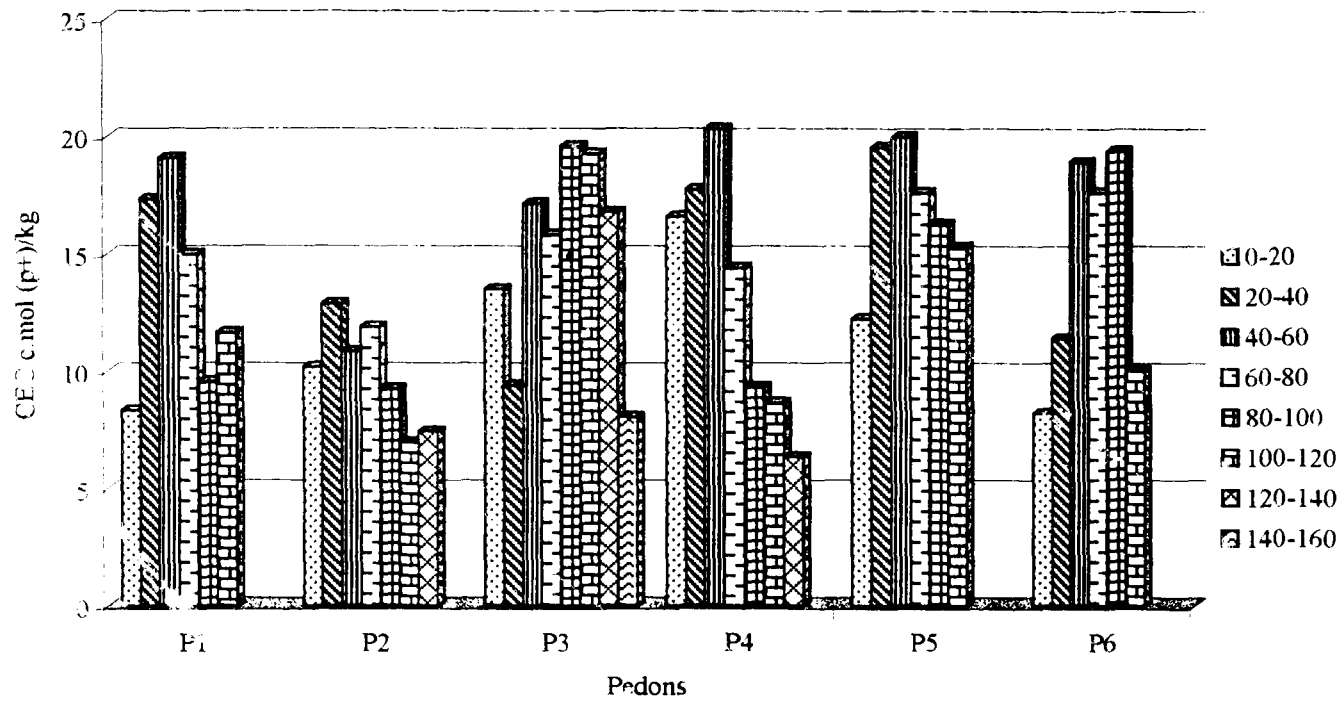


Fig.13.1: Charge characteristics - Cation exchange capacities of Pedons

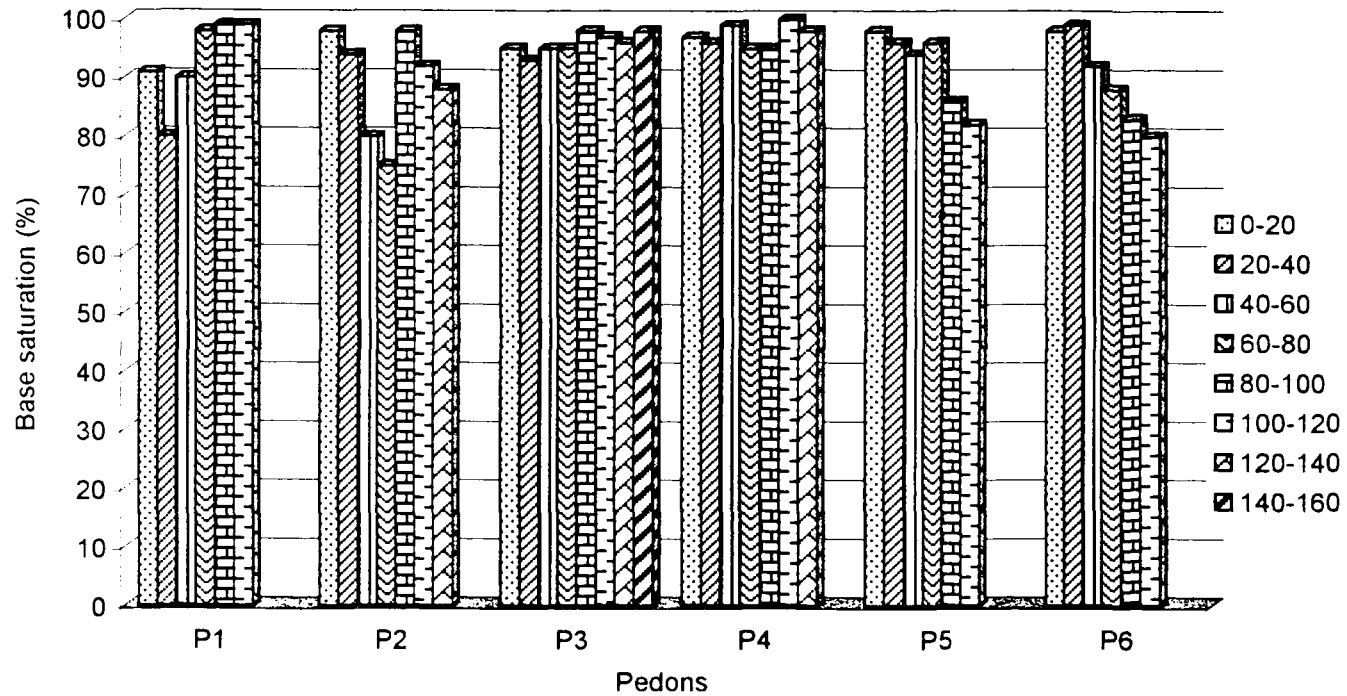


Fig.13.2: Charge characteristics - Per cent base saturation of Pedons

obtained by Walia and Chamuah (1988). In all the pedons calcium was noted to be dominant among the exchangeable cations followed by magnesium throughout the depth, some irregular changes in concentrations of magnesium were also been noticed due to unexplained reasons. Base saturation in all the pedons (Fig. 13.2) was found to fluctuate from medium to high in the soil, which may be attributed to the medium to high amount of rainfall recorded in the study area.

5.4 Classification of soils

Red soils of the tropics encompass a range of soils, which in modern soil classification system can be grouped under Entisols, Inceptisols, Alfisols, Ultisols, Mollisols and Oxisols. In well drained environments in tropics, weathering and soil formation frequently lead to the formation of deep red soils. The actual colour of the soil is a function of the soil moisture regime, soil temperature regime, the nature of parent martial (particularly iron content) and the physiographic position of the soil. Depending upon the degree of influence of the factors indicated above, the colour of the soil varied from reddish brown to dusky red and consequently the term red soils.

As cited above red soils may belong to many orders in Soil Taxonomy (Soil Survey Staff, 1998). Eswaran (1988) defined red soils as "Those soils that have a hue of 5YR or redder and have a CEC (1 NNH_4OAc at pH 7) of 24 meq. per 100g clay or less or an ECEC of 18 meq per 100 g clay or less. They may have an argillic horizon or oxic horizon but does not have a

cambic horizon and they have a thermic, isothermic or warmer iso-temperature regimes. These soils may have a range of diagnostic horizons and features and include the following.

- Argillic
- Kandic
- Oxic
- Plinthite
- Petroplinthite
- Petroferric horizon

Based on the morphological, physical and chemical characteristics of the six selected pedon, the pedons can be classified under USDA classification as indicated below:

Pedon 1: Fine, mixed, isohyperthermic family of Typic Haplustalfs

The particle size class in the control section of the pedon is clayey, it is grouped as fine.

The medium cation exchange capacity of the clay which ranges from 17.0 cmol (p^+) kg^{-1} to 38.0 cmol (p^+) kg^{-1} and slightly acidic nature of the soil indicate the presence of both Kaolinite type of clay minerals and montomorillonite/illite type of clay particles in the soil, thereby qualifying to be grouped under mixed type of mineralogy.

The mean annual soil temperature range of the study area is 22°C or higher, so the temperature regime is isohyperthermic. The soils of this pedon donot have plaggen epipedon and have an argillic horizon, thereby classified as Alfisols at order level of classification.

Since these soils have Ustic moisture regime (soils of Ustic moisture regime remain dry in some of all parts for 90 days or more cumulative days in normal years and moist in some part either for more than 180 cumulative days per year or for 90 or more consecutive days), they are classified as Ustalfs at suborder level. Soils of this pedon donot have the properties of any of the great groups of Ustalfs viz., Durustalfs, Plinthustalfs, Natrustalfs, Kandiustalfs, Kanhaplustalfs, Paleustalfs, Rhodustalfs, therefore, these soils belong to Haplustalfs great group.

At subgroup level of classification, they are classified as Typic Haplustalfs as soils of this pedon, donot satisfy the requirements of any of the sub groups of Haplustalfs viz., Lithic Haplustalfs, Aquertic Haplustalfs, Udertic Haplustalfs, etc.

Pedon II: Fine, Kaolinitic, isohyperthermic family of Kandic Paleustalfs.

Like pedon I, the particle size class in the control section of the pedon is also clayey, it is grouped as fine; the medium cation exchange capacity of the clay ranges from 17 cmol (p^+) kg^{-1} (in BC horizon) to 31.0 cmol (p^+) kg^{-1} (in Ap horizon) and near neutral soil reaction indicates the dominance of Kaolinite (1:1, non-expanding type of clay minerals) mineral in the soil, thereby it is

grouped as kaolinitic type of mineralogy: the mean annual soil temperature range of the study area is 22° or higher hence soil temperature regime is isohyperthermic. These soils belong to order Alfisols, at order level of classification as per cent base saturation is more than 35 per cent in major part of the argillic horizon.

At suborder level of classification, they belong to Ustalfs as pedons samples were collected from an area having Ustic moisture regime.

At great group level, these soils are classified as paleustalfs, as these soils have a clayey particle size class throughout. There is increase in clay content upto 100 cm and reaches its maximum (52.1%). Thereafter, it decreases which is not more than 20 per cent.

As these soils meet the requirements of Kandic properties i.e., Paleustalfs that have a CEC of less than 24 C.mol (p^+) kg^{-1} clay in 50 per cent or more. Either of the argillic horizon, hence the term Kandic Paleustalfs.

Pedon III: Fine, mixed, isohyperthermic family of Typic Haplustepts

The soils of this pedon are grouped under fine, as particle size class in the control section of pedon was found to be clayey; mixed, as cation exchange capacity values varied between 46.0 cmol (p^+) kg^{-1} and 24.0 cmol (p^+) kg^{-1} and soil reaction was slightly acidic indicating presence of 1 : 1 Kaolinite type and 2 : 1 montomorillinite type of clay minerals; isohyperthermic, as mean annual soil temperature is 22°C or higher.

At order level of classification, these soils belong to Inceptisols as, a cambic horizon with its upper boundary within 100 cm of the mineral soil surface and its lower boundary at a depth of 25 cm or more below the mineral soil surface.

The pedon samples were collected from the area having Ustic moisture regime hence at suborder level they are grouped as Ustepts.

At great group level of classification, the soils of this pedon donot qualify for great groups viz., Durusteps, Calciustepts and Dystrustepts therefore classified as Haplustepts.

Since soils of this pedon donot have the characteristics of any of subgroups of great group Haplustepts viz., Lithic Haplustestepts, agric Haplustepts, vertic Haplustepts, etc., they are classified as Typic Haplustepts.

Pedon IV: Fine, Mixed, isohyperthermic family of Typic Haplustalfs.

Soils of this pedon are having high clay content hence qualifying to be grouped under fine.

Since these soils have both the Kaolinite (1 : 1) type of clay mineral and 2 : 1 type montomorillonite/illite type of clay mineral they are classified as mixed.

The mean annual soil temperature of the study area is 22°C or higher consequently the term isohyperthermic.

At order level of classification soils of this pedon qualify for Alfisols as per cent base saturation of the soils in major part of the argillic horizon is more than 35 per cent. B, these soils donot have a plaggen epipedon and have an argillic horizon.

The soil moisture regime of the study area is Ustic; hence at suborder level of classification they are referred as Ustalfs.

They belong to Haplustalfs great group as necessiates of any of the great groups of Ustalfs; these soils are grouped as Haplustalfs at great group level of classification.

At sub group level of classification, soils of this pedon are classified as Typic Haplustalfs as they donot have any of the characteristics of Haplustalfs subgroups

Pedon V: Fine, mixed, isohyperthermic family of Typic Haplustalfs.

The soils of this pedon are having high clay content they are grouped as fine; both Kaolinite (1:1) type of clay minerals and montmorillonite or illite (2:1) type of clay minerals are present, the mineralogy is mixed; the mean annual soil temperature of study area is 22°C or higher, the soil temperature regime is isohyperthermic.

At order level of classification, soils of this pedon belong to Alfisols as there was no plaggen epipedon, have an argillic horizon and per cent base saturation (by sum of cations) of more than 35 per cent.

Since these soils are collected from an area having Ustic moisture regime, at suborder level they belong to Ustalfs.

At great group level of classification, they belong to Haplustalfs as these soils donot fulfill the requirements of any of the great groups of Ustalfs.

At sub group level of classification, these soils belong to Typic Haplustalfs as they donot have the necessiates of any of the sub groups of Haplustalfs.

Pedon VI: Fine, mixed, isohyperthermic family of Typic Paleustalfs

The soils of pedon VI are having high clay content hence these soils are grouped under fine; both Kaolinite and montmorillonite/illite clay minerals are prevalent in the soils of this pedon, the mineralogy is mixed; the soil temperature regime is isohyperthermic mean annual soil temperature of study area is 22°C or higher.

The soils of this pedon, at order level of classification, classified as Alfisols because of the reasons cited below:

A. The per cent base saturation of the soils in major part of the argillic horizon is more than 35 per cent.
B. These soils do not have a plaggeon epipedon, and
C. Have an argillic horizon

The Ustic moisture regime is the soil moisture regime of the study area hence at suborder level of classification they are classified as Ustalfs.

The Pleustalfs is the great group to which soils of this pedon belong to, because –

A. There was no densic, Lithic or paralithic contact within 150 cm of the mineral soil surface and an argillic horizon that has both (a) within 150 cm of mineral soil surface either (i) with increasing depth, no clay decrease of 20 per cent or more from the maximum clay content. (iii) 5 per cent or more (by volume) skeletons on faces of pedons in the layer that has a 20 per cent lower clay content.

At sub group level of classification, soils of this pedon are classified as Typic Paleustalfs as these soils do not possess adequate or necessary requirements of other subgroups of Paleustalfs.

5.5 Available nutrient status of soils of the study area

5.5.1 Available nitrogen

The examination of data on the available nitrogen content (Table 6) of all the pedons varied between low and medium throughout the depth. However, available nitrogen content was found to be maximum in the surface horizon and regularly decreased down the profile. Similar observations

were made by Parasuraman and Jayaraj (1982). The reason that could be attributed to maximum available nitrogen content present at the surface horizon is mainly because of the fact that cultivation of crops are mainly confined to the surface horizon (0-18 or 0-20 cm rhizosphese) only at regular intervals and nitrogen content is supplemented by the external application of fertilizer during crop cultivation.. As the available nitrogen content varied between low to medium, there remains always a necessity to supplement the deficient nitrogen by the external source i.e., through fertilizers.

5.5..2 Available phosphorus

The data on available phosphorus (Table 6) indicated that available phosphorus content of all pedons varied between low to medium. However highest phosphorus content started accumulating next to surface horizon and decreased regularly with depth. Similar observations were obtained by Parasuraman and Jayaraj (1982) and Mruthunjaya (1991). The reason that could be attributed to such observations may possibly be due to the confinement of crop cultivation to the rhizosphere and 'P' fixation by Fe (Mathan, 1990) and supplementing the depleted phosphorus by fertilizers. However, by looking at the available phosphorus content in the soil, there is always necessity to supplement the deficient phosphorus content in the soil for the better crop yields.

5.5.3 Available potassium

The examination of data on the available potassium content (Table 6) of the soils in all the pedons revealed that the available potassium content varied from low to medium. Highest available potassium content was noticed in the surface horizon, which decreased regularly down the profile. These observations are in accordance with the observations made by Parasuram and Jayaraj (1982) and Mruthunjaya (1991). The reason attributed to such observation may possibly be due to the fact that cultivation of crops remained confined to the rhizosphere mainly and the depleted potassium in the surface horizon was replenished by external application i.e., fertilizers during crop cultivation. However, by visualizing the available potassium content of the soil, there is necessity to supplement the deficient potassium content in the soil by external source for the better augmentation of the crop production.

5.5.4 Available micronutrients

The examination of data on available micronutrients of the soils presented in Table 6 revealed that DTPA extractable iron, zinc and manganese content varied from deficient to optimum in the pedons. Copper content of soils was found to be deficient in all the pedons. The DTPA extractable iron content of soils showed a decreasing trend with the depth of the pedon. This was observed in all the pedons and in accordance with the similar observation made by Mehta *et al.* (1964), Shukla and Singh (1973) respectively. The DTPA extractable copper content of soils in all the pedons showed a regular decrease from surface to subsurface horizons. Similar views were expected by

Kavimandan *et al.* (1964) and Bansal *et al.* (1969). The DTPA extractable manganese content of soils in all the pedons was noted to decrease regularly throughout the pedon. Similar trend was also observed by Mehta *et al.* (1964) and Shukla and Singh (1973).

5.5.5.1 Improved cropping and cropping system

A sizeable area of red soils is cultivated under rainfed situations. With the adoption of suitable management practices and careful planning the production under dryland conditions can be doubled. The main cause for low yields is due to erratic rainfall in marginal lands. These lands can be effectively be used for production fodder, fuel, fibre and fruits. Some fruit trees such as mango, tamarind, Tizypes, jack, custard apple have already been cultivated in these areas in recent years. However mango is highly preferred by the dryland farmers.

Since the present area of investigation like many other parts of the rainfed soil regions of the state receives bimodal rainfall spread over 5-6 months (April-May and July-October), it becomes quite obvious that such areas with deep soils have a potential for double cropping by selection of suitable crops and timely completion of operations. Some of the recommended crops for double cropping systems based on rainfall distribution are:

Cowpea (Variety: KBC-1) (April-May)	Ragi (varieties: Indaf 5 & 9) (August)
Ground nut (Varity: JL24) (May)	Ragi (transplanted) (September)
Jowar (Variety CSH-5) (April)	Horsegram or Ragi (Transplanted) (Variety: KBH-1) (August)
Sesamum (Variety: TMV-3) (April)	Horsegram or Ragi (Transplanted) (Variety: KBH-1) (August)
Sunflower (Variety: KBSH-1) (May)	Ragi (Transplanted) (August)

5.5.5.2 Nutrient management

Continuous runoff and soil erosion coupled with nutrient loss in the study area has made crops not only thin but also hungry which is seen from the chemical analysis of the pedons, as indicated by the available nutrient status of the soils. Hence, there is a need to replenish the fertility loss for sustaining productivity. Farmyard manure and chemical fertilizers improve the physical properties of soils, encourage biological activity and meet the immediate requirements of plant nutrients. Therefore, combined application of both organic manures and recommended dose of fertilizers to enhance the crop yield and improve physico-chemical and biological properties of soil.

Use of low cost fertilizers and bio-culture has the benefit of reduced cost and increase in yield. Use of legumes help in fertility build up in neutral and slightly acidic soils. Band placement of fertilizer using seed cum fertilizer drill and split application (2-3 doses) of recommended nitrogen in coarse textured soil increase its use efficiency, use of soil amendments like

lime in acidic soil and phosphate fertilizer in 'P' deficit soil improves soil productivity.

5.5.5. 3 Harvesting runoff water by constructing farm pond

These soils are prone to crusting and compaction, which is caused by high intensity raindrops. It not only hampers the emergence of the seedlings of small millets, but also causes a higher rate of runoff. Therefore a part of inevitable runoff from lands can be harvested in small farm ponds excavated at suitable points. The stored water in these ponds can be used to give a protective irrigation to one fourth or one third area of land during long breaks in monsoon. A pond size of 250 m^3 capacity is recommended for catchment of one hectare receiving an annual rainfall of 750 mm. These ponds can be lined with 1 : 8 cement and soil to reduce seepage loss which is normally high in red soils.

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SUMMARY

VI. SUMMARY

Soils are the most vital component for providing food and fibre to meet the increasing requirements of ever-growing population in the country. They need to be used rationally and conserved properly to ensure increased agricultural production on sustainable basis and to leave a better heritage for posterity. Hence knowledge about the potentials and problems of different soils, their extent and distribution is essential for optimal and sustainable use of this limited resource. Information on soil properties (physico-chemical) is of greater importance for better choice of alternative land uses. In this context the study entitled "Characterisation and classification of Major soils of Institute Village Linkage Programme Area, Doddaballapura Taluk, Bangalore Rural District" was undertaken to assess the characteristics, potentialities and constraints of the soils occurring in the study area under different topographical conditions for better management of soil resources for sustained crop production.

Six soil profiles representing Institute Village Linkage Programme Area were selected for the present investigation after thorough survey of study area. The soils were characterised in the field for their morphometric properties and later in the laboratory for their physico-chemical and nutritional characteristics. The results of the investigations are summarised below:

1. Depth of all the profiles of the study area was very deep.

2. Colour of the soil varied from dark reddish brown to dusky red or red colour

and found to be influenced mainly by the type of parent material, low organic carbon content, warmer temperature regime (isohyperthermic) and dry moisture regime (Ustic) prevalent in the area.

3. The common structure found in all the profiles is sub-angular blocky both at surface and subsurface except in pedon III where in subsurface horizons massive wedge shaped aggregates were present.

4. Presence of ochric epipedon (pale coloured surface horizon) and argillic horizon in subsurface horizons indicated translocation of clay and its deposition in the subsurface horizons. Chemical weathering was found to be highly pronounced in all the pedons.

5. The texture of the soils was found to vary between sandy clay loam to clay, however, clay content increased down the profile with simultaneous decrease in sand content indicating illuviation of clay in the soil.

6. Soil reaction of all the soils were found to be between near neutral and slightly acidic, which may be attributed to the presence of 1 : 1 type of kaolinite clay (dominant clay group), low bases and low organic carbon content in the soil.

7. The electrical conductivity of all the pedons were very low indicating less amount of salt present in the soils.

8. The organic carbon content of all the pedons was found to vary between low to medium. This is due to low vegetative cover, pronounced soil erosion and

warmer climate leading to less accumulation of organic carbon in the soils of the study area.

9. The cation exchange capacity of the soils varied between low to medium due to the predominance of (1 : 1) kaolinite clay and low organic matter (organic carbon) content.

10. In all the pedons calcium was found to be the most dominant cation followed by magnesium throughout the depth of the profile. Base saturation of soils was found to fluctuate between medium and high in the soil.

11. The available nitrogen, phosphorus and potassium content of the soils were found to vary between low (subsurface horizons) to medium (surface horizons) respectively.

12. The available Micronutrients of the soils i.e., DTPA extractable iron, zinc, copper and manganese content varied from deficiency to optimum.

13. Soils of all the six pedons were classified under two main orders i.e., Alfisols, Inceptisols.

14. Based on the above physico-chemical properties of the soils, proper management practices that will help in soil and moisture conservation to enhance the present crop production level can be attempted to maintain the sustainability of the existing natural resources of the area.

REFERENCES

VII. REFERENCES

Arunprasad, Totey,N.G., Singh,A.K., Kulkarni,R., Khatri,P.K., Bhownik,A.K., Chouhan,J.S. and Dahia,V.J.K., 1989, Profile development in relation to topography. *J.Indian Soc.Soil Sci.*, **37** : 790-797.

Badrinath,M.S., Krishnappa,A.M., Sridharherle,P., Patil,B.N., Balakrishna Rao,K. and Janardhana Gowda,N.A., 1987, Distribution of available micronutrients in saline soils of Ankola (Uttar Kannada), Coastal Karnataka, India. *Curr.Res.*, **26** : 24-26.

Bhadrapur,T.G. and Seshgiri Rao,T.V., 1977, Salt affected black Soils, chemical characteristics and sodium divalent cation exchange relationship. *J.Indian Soc.Soil Sci.*, **25** (1) : 11-17.

Bhaskar,B.P. and Subbaiah,G.V., 1995, Genesis, characteristics and classification of laterites and associated soils along east coast of Andhra Pradesh. *J.Indian Soc.Soil Sci.*, **43** (1) : 107-112.

Black,C.A., 1965, *Methods of soil analysis part II.* Agronomy monograph No.9, American Society Agronomy, Madison, Wisconsin.

Das,P.K., Nath,S. and Banerjee,S.K., 1988, Characteristics of the two soils of sub alpine region in eastern Himalayas. *J.Indian Soc.Soil Sci.*, **36** : 576-580.

Das,P.K., Sahu,G.C. and Nanda,S.S.K., 1992, Nature of acidity and exchange chemistry of some lateritic soils of Orissa. *J.Indian Soc.Soil Sci.*, **40** (3) : 425-430.

Dasai,A.D., 1942, The nature and relationship of black cotton soils and red earths of Hyderabad. *Hyderabad Dept. Agric.Bull.* No.10, Hyderabad State.

Dasog,G.S. and Hadimani,A.S., 1980, Genesis and chemical properties of some vertisols. *J.Indian Soc.Soil Sci.*, **28** (1) : 49-56.

Datta,S. and Karmakar, 1995, Characterization and classification of Soils of Instructional-cum-Research Farm of Assam Agricultural University, Jorhat. *J.Indian Sci.Soc of North-East India.* **8** (1) : 40-46.

Diwakar,D.P.S. and Singh,R.N., 1994, Characterization of soils of diara land occurring in Gangetic plains of Bihar. *J.Indian Soc.Soil Sci.*, **42** (3) : 420-425.

Elahi,S.F., Hossain,M.F. and Kamal,A.S.M.M., 1996, Characteristics of some soils developed on madhupura clay in Bangladesh. *J.Indian Soc.Soil Sci.*, **44** (3) : 482-488.

Esu,I.E. and Ojanuga,A.G., 1985, Morphological, physical and chemical characteristics of Alfisol in Kaduna Area of Nigeria.

Eswaran,H. and Wang Chaw Bin, 1978, A study of deep weathering profile on granite on peninsular Malaysia. I. Physico chemical and micromorphological properties. *Soil Sci.Am.J.*, **42** : 144-140.

Eswaran,H., 1988, Taxonomy and management related properties of the red soils of Africa. *Proc.Int.Symp.* Red soils Harare, Zimbabwe, 1986; Publ. IRDC-MR **170e** : 1-23.

F.A.O., 1993, Frame work for Land Evaluation, *Soils Bulletin*, **32**, Rome.

Gangappa,P.V., 1989, Nature of salt application in salt affected soils of Challakere taluk, Chitradurga district. M.Sc. (Agri.) Thesis, Univ. Agril.Sci., Bangalore.

Govindarajan,S.V. and Datta Biswas,N.R., 1968, Characterization of certain soils in the subtropical humid zone in the south eastern part of Indian soils of Muchkand basin. *J.Indian Soc.Soil Sci.*, **16 :** 117-186.

Gupta,R.D. and Tripathi,B.R., 1993, Characterization of rice growing and adjacent virgin soils. I. Morphology, properties, genesis and classification. *J.Indian Soc.Soil Sci.*, **41** (4) : 720-724.

Jackson,M.L., 1973, *Soil chemical analysis*, Prentice Hall of India (Pvt.) Ltd., New Delhi.

Kaista,B.P. and Gupta,R.D., 1993, Morphology, minerology genesis and classification of soils of the subhumid temperature high lands of the central Himalayas. *J.Indian Soc.Soil Sci.*, **40** (1) : 120-124.

Katyal,J.C. and Sharma,B.D., 1991, DTPA extractable and total Zn, Cu, Mn and Fe in Indian soils and their association with some soil properties. *Geoderma*, **49** : 165-179.

Krishnamoorthy,P. and Govinda Rajan,S.V., 1977, Genesis and classification of associated red and black soils under Rajolibunda diversion irrigation scheme (Andhra Pradesh). *J.Indian Soc.Soil Sci.*, **25** (3) : 239-246.

Krishnan,P., 1993, Physico-chemical properties and classification of some red and lateritic soils of southern India. NBBS Publ., **37**pp 57.

Mruthunjaya,S., 1991, Characterization of some salt affected soils of Cauvery and Vanivilas Sagar Command areas of Karnataka, M.Sc. (Agri.) Thesis, Univ.Agril.Sci., Bangalore.

Murthy,R.S., Hirekerur,L.R., Deshpande,P.B. and Venkata Rao,B.V., 1982, Benchmark soils of India, National Bureau of Soil Survey and Land Use Planning, Nagpur, p 374.

Naidu,L.G.K., Reddy,R.S., Raja,E., Shivaprasad,C.R. and Krishnan, p.1994. Characterization of Mango (Mangifera-Indica) Growing Soils and Soil Related Constraints. *Indian J.Agric.Sci.*, **64** (6) : 359-363.

Narayana Rao,K., Dasog,G.S. and Satyanarayana,T., 1993, Characterization and Classification of some Lateritic soils of North Karnataka. NBSS Publ. **37** pp 39.

Nirmalya Bala and Sahu,G.C., 1993, Characterization and classification of soils of hill slope of middle Andaman Island. *J.Indian Soc.Soil Sci.*, **41 :** 133-137.

Pal,D.K., Nath,S., Banerjee,S.K. and Sharma,S.K., 1985, Characteristics of some forest soils of Darjeeling. Himalayan Region under pinus Patula. *J.Indian Soc.Soil Sci.*, **33 :** 84-95.

Parasuraman,S. and Jayaraj,S., 1982, Distribution of Total and available N, P, and K of the Major soil series of profiles of Black soils of Coimbatore District, Tamilnadu, *Madras Agric.J.*, **69** : 825-829.

Parvathappa,H.C., 1964, Study of Characteristics and Productivity of Red Soils of Mysore State, M.Sc. (Agri.) Thesis. Faculty of Soil Science, Regional Postgraduate Training Centre, Agricultural College and Research Institute, Coimbatore.

Piper,C.S., 1950, *Soil and plant analysis*. Hans Publ., Bombay..

Piper,C.S., 1950, *Soil and plant analysis*. Inter Sci. Pub., In. Proc.Int.Symp. New York.

Prakash,N.B., Kenchanagowda,S.K. and Siddaramappa,R., 1993, Phosphorus Adsorption and desorption in Alfisols of Karnataka. NBSS Publ. **37** pp 205.

Rajamannar,A. and Krishnamoorthy,K.K., 1978, A note on the influence of altitude on the physico-chemical characters of Forest soils. *J.Indian Soc.Soil Sci.*, **26** (4) : 399-400.

Raychaudari,S.P., 1963, Report on Land Resource in India – Committee on Natural Resources, Planning Commission, Govt. of India.

Raychaudhari,S.P., 1941, Studies on the physico-chemical properties of Associated Black and Red soils of Nyasaland Protectorate, British Central Africa. *J.Indian Soc.Soil Sci.*, **11** : 100-109.

Reddy,D.R., Vasudeva Rao,A.E. and Raghumohan,N.G., 1993, Morphology *and physico-chemical properties of Red soils (Alfisols) under

irrigated and unirrigated conditions of Nagarjunasagara Project Area of Andhra Pradesh. Red and Lateritic soils of India-Resource Appraisal and Management. NBSS Publ., **37** : 106-110.

Reddy,P.S.A.,Nagabhushana,S.R.,Krishnan,P., Natarajan,A., Shivaprasad,C.R., Venugopal,K.R., Prabhakara and Urs,M.S.D.A., 1993, Distribution, characterization and classification of Red and Lateritic soils of India. NBSS Publ **37** pp 15.

Roy,B.B. and Landey,R.T., 1962, Studies of Red and Lateritic soils of Mord water shed area of Raigarh district. *J.Indian Soc.Soil Sci.*, **32** (6) : 294-302.

Rudramurthy,H.V., Dasog,G.V. and Vasuki,N., 1996, Genesis and Classification of Associated Red and Black Pedons of North Karnataka. *Mysore J.Agric.Sci.*, **30** (4) : 321-330.

Sahu,G.C. and Mishra,K.N., 1997, Morphology, Characteristics and Classification of soils of an irrigated river flood plain in the eastern coastal region. *J.Indian Soc.Soil Sci.*, **45** (1) : 152-156.

Sahu,G.C., Patnaik,S.N. and Das,P.K., 1990, Morphology, genesis, mineralogy and classification of soils of Northern Plateau Zone of Orissa. *J.Indian Soc.Soil Sci.*, **38** : 116.

Sakal,R., Singh,A.P. and Singh,S.P., 1988, Distribution of Available Zinc, Copper, Iron and Manganese in old alluvial soils as related to certain soil characteristics. *J.Indian Soc.Soil Sci.*, **36** : 59-63.

Sakal,R., Singh,B.P., Singh,A.P., Sinha,R.B., Singh,S.P. and Tha,S.N., 1986, Availability of Zinc, Iron, Copper, Manganese in sub-Himalayan chill and forest soils as Influenced by Certain soil properties. *J.Indian Soc.Soil Sci.*, **34** : 191-193.

Sarma,P.K., Bordoli,P.K., Baruah,T.C., Das,K.N. and Borah,N., 1997, Physical properties of Two soils (Alfisols and Inceptisols) from different Command areas of Assam. *J.Indian Soc.Soil Sci.*, **45** (4) : 709-712.

Sarma,V.A.K., Krishnan,P. and Budihal,S.L., 1987, *Soil resource mapping of different states in India*: Laboratory methods, NBSS Publ., 49 pp.

Satish,G.C., 1991, Physical and chemical properties of soils of different agricultural situations in Dakshina Kannada district of Karnataka. M.Sc. (Agri.) Thesis, submitted to University of Agricultural Sciences, Bangalore.

Sehgal,J., Blum,W.E. and Gajbhiye,K.S., 1998, Red and lateritic soils: Red and lateritic soils of the world, Vol.1 : 15 pp.

Sen,T.K., Dubey,P.N. and Chamuah,G.S., 1997, Characteristics and classification of some soils of Barak valley in Assam. *J.Indian Soc.Soil Sci.*, **44** (3) : 470-475.

Sen,T.K., Nayak,D.C., Maji,A.K. and Chamuah,G.C., 1993, Pedogenic characteristics of some red soils of Manipura, NBSS Publ., **37**pp62.

Sharma,B.K. and Rao,K.S.K., 1957, Morphological physico-chemical and chemical characteristics of apple growing soils of Himachal Pradesh. *J.Indian Soc.Soil Sci.*, **5 :** 223-229.

Sharma,P.D. and Abdul,Q. Qaher, 1989, Characterization of some Himalayan protected and eroded forest soils. *J.Indian Soc.Soil Sci.*, **37 :** 113-120.

Shivaprasad,C.R., Reddy,R.S., Sehgal,J and Velayutham,M., 1998, Soils of Karnataka for optimizing Land use. NBSS Publ., **47**pp 15.

Shivaprasad,C.R., Reddy,R.S., Sehgal,J. and Velayutham,M., 1998, Soils of Karnataka for optimizing Land use. NBSS Publ. **47** pp 35.

Shrikanth,N.D. and Bapat,M.V., 1993, Characterization and classification of soils in relation to different parent rocks and land forms. *J.Indian Soc.Soil Sci.*, **41** (2) : 326-330.

Singh,R.A., 1980, *A text book of soil physical analysis*. Kalyani Publishers, New Delhi.

Sivasankaran,K., Mithyantha,M.S., Natesan,S. and Subbarayappa,C.T., 1993, Physico-chemical properties and nutrient management of red and lateritic soils under plantation crops in Southern India. NBSS Publ. **37** pp 280.

Soil Survey Staff, 1998, keys to soil taxonomy. *Agree Handb.*., U.S.Dep., Agric., 436 pp.

Venkatesh,M.S. and Sathyanarayana,T., 1994, Status and distribution of potassium in vertisols of North Karnataka. *J.Indian Soc.Soil Sci.*, **42** (2) : 229-233.

Walia,C.S. and Rao,Y.S., 1996, Genesis, characteristics and Taxonomic classification of some Red soils in Bundelkhand Region of Uttarpradesh. *J.Indian Soc.Soil Sci.*, **44** (3) : 576-581.

Yassoglou,N., Kosmas,C., Moustakas,N., Mermut,A.R., Yaalon,D.H. and Kapoor,S., 1997, The Red soils, their origin, properties, use and management in Greece. Special Issue Red Mid-terrain soils meeting, Adama Turkey **28** (3-4) : 261-278.

Zende,N.A., 1987, Soils of Nagaland and Relation to physiography, their characteristics and classification. *J.Indian Soc.Soil Sci.*, **35** : 706-711.

APPENDIX

Appendix I : Critical limits of available nutrients (essential)

Sl. No	Element	Low	Medium	High
1	Available N (kg/ha)	<280	280-560	>560
2	Available P (kg/ha)	<10	10-24.6	>24.6
3	Available K (kg/ha)	<118 •	118-280	>280

Appendix II : Critical limits for DTPA extractable micronutrients (Zn, Cu, Mn & Fe)

Sl. No	Element	Low	Marginal	Adequate
1	Zinc (ppm)	0.5	0.5 – 1.00	1.00
2	Copper (ppm)	<0.2	-	>0.20
3	Manganese (ppm)	<1.0	-	>1.00
4	Iron (ppm)	2.5	2.5 – 4.00	4.00